Lecture Notes 2025

Notes for Lazarsfeld's Positivity in Algebraic Geometry

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Contents

Part I

Background and Basic Concepets

Chapter 1 Locally free sheaves

The most important examples of quasi-coherent sheaves are the locally free sheaves. As the name suggests, these are sheaves which are locally isomorphic to a direct sum of copies of the structure sheaf of the scheme. Because of this 'freeness' property, these sheaves are in many respects the nicest examples of sheaves on a scheme and the easiest to work with. They are also the algebraic counterpart to the vector bundles in topology.

1.1 Locally free sheaves

Definition 1.1.1 (Locally free sheaves)

Let X be a scheme. We say that an $\mathscr{O}_X$-module $\mathscr{E}$ is ***free*** *if it is isomorphic to a direct sum of copies of $\mathscr{O}_X$ and it if* ***locally free*** *if there is an open cover $\{U_i\}_{i\in I}$ of X such that the restriction $\mathscr{E}|_{U_i}$ is free for every i. The cover $\{U_i\}$ is called a* ***trivializing cover****, and a locally fee sheaf which is globally free is sometimes said to be* ***trivial****.*

Definition 1.1.2 (Rank)

If $P \in X$ is a point, say $P \in U_i$, the ***rank*** *of $\mathscr{E}$ at P is the number of copies of $\mathscr{O}_{U_i}$ need to express $\mathscr{E}|_{U_i}$ as a direct sum of $\mathscr{O}_{U_i}$'s.*

If $\mathscr{E}$ has rank equal to r at every point of X, we say that $\mathscr{E}$ is locally free of rank r and write $\operatorname{rank}\mathscr{E} = r$.

Remark

(1) The rank may be finite or infinite, but we will almost exclusively concern ourself with the case of finite rank

(2) A locally free sheaf $\mathscr{E}$ is allowed to have different ranks at different points at X, but the rank will be constant on each connected component of X.

The rank is perhaps better viewed as a function $X \to \{0, 1, 2, \ldots\} \cup \{\infty\}$. To discuss this function precisely, we make the following definition, which applies to any coherent sheaf.

Definition 1.1.3 (Fiber of coherent sheaf at a point)

Let $\mathscr{E}$ be a coherent sheaf and let $P \in X$ be a point. The ***fiber of $\mathscr{E}$ at P*** *is defined as the $\kappa(P)$-vector space*

$$\mathscr{E}(P) = \mathscr{E}_P/\mathfrak{m}_P\mathscr{E}_P.$$

If $U \subseteq X$ is an open subset containing P and $s \in \Gamma(U, \mathscr{E})$ is a section of $\mathscr{E}$ over U, we denote by $s(P)$ the image of the germ $s_P \in \mathscr{E}_P$ in the fiber $\mathscr{E}(P)$.

Remark Geometrically, $\mathscr{E}(P) \cong \mathscr{E}_P \otimes_{\mathscr{O}_{X,P}} \kappa(P)$ is the $\kappa(P)$-vector space which corresponds to the pullback of $\mathscr{E}$ via the map $\iota : \operatorname{Spec}\kappa(P) \to X$.

Clearly, if $\mathscr{E}$ is locally free of rank r, then $\mathscr{E}_P$ is a free $\mathscr{O}_{X,P}$-module of rank r and $\mathscr{E}(P)$ is a free $\kappa(P)$-vector space of dimension r at every point P.

Proposition 1.1.1

Let X be an integral Noetherian scheme and let $\mathscr{E}$ be a coherent sheaf. The following are equivalent:

(i) $\mathscr{E}$ is locally free.

(ii) $\mathscr{E}_P$ is a free $\mathscr{O}_{X,P}$-module for every $P \in X$.

(iii) The function $P \mapsto \dim_{\kappa(P)} \mathscr{E}(P)$ is a constant.

The next proposition summaries some of the basic properties of locally free sheaves.

Proposition 1.1.2

Let X be a scheme and let $\mathscr{E}$ and $\mathscr{F}$ be two locally free $\mathscr{O}_X$-modules of rank e and rank f respectively. Then

(i) The direct sum $\mathscr{E} \oplus \mathscr{F}$ is locally free of rank $e + f$.

(ii) The tensor product $\mathscr{E} \otimes_{\mathscr{O}_X} \mathscr{F}$ is locally free of rank $e \cdot f$.

(iii) The Hom sheaf $\mathscr{H}om_{\mathscr{O}_X}(\mathscr{E}, \mathscr{F})$ is locally free of rank $e \cdot f$.

(iv) The dual sheaf $\mathscr{E}^\vee = \mathscr{H}om_{\mathscr{O}_X}(\mathscr{E}, \mathscr{O}_X)$ is locally free of rank e. The double dual $(\mathscr{E}^\vee)^\vee$ is canonically isomorphic to $\mathscr{E}$. Moreover, there is a canonical isomorphism

$$\mathscr{E}^\vee \otimes_{\mathscr{O}_X} \mathscr{F} \xrightarrow{\sim} \mathscr{H}om_{\mathscr{O}_X}(\mathscr{E}, \mathscr{F}).$$

Example 1.1 (The integers) Let $X = \operatorname{Spec} \mathbb{Z}$. If $\mathscr{E}$ is an $\mathscr{O}_X$-module of finite type, then $\mathscr{E} = \widetilde{M}$ for some finitely generated $\mathbb{Z}$-module M, and by the structure theorem for f.g. abelian groups, we may write $M = \mathbb{Z}^r \oplus \operatorname{Tor}(M)$, where $\operatorname{Tor}(M)$ is a finite direct product of groups of the form $\mathbb{Z}/n\mathbb{Z}$. If $\mathscr{E}$ in addition is required to be locally free, it must hold that $\operatorname{Tor}(M) = 0$ (otherwise, if p is a prime factor of an n appearing in one of the summands of $\operatorname{Tor}(M)$, the stalks at (p) will not be free). Hence $\mathscr{E} = \widetilde{\mathbb{Z}^r} = \mathscr{O}_X^r$, and we conclude that every locally free sheaf on $\operatorname{Spec} \mathbb{Z}$ is trivial.

Example 1.2 (The affine line) The argument of the Example 1.1 in fact applies over any PID A. Every $\mathscr{O}_X$-module of finite type on $X = \operatorname{Spec} A$ must have the form $\widetilde{M}$ for $M = A^r \oplus \operatorname{Tor}(M)$ where $\operatorname{Tor}(M)$ is a f.g. torsion module, and if we require $\widetilde{M}$ to be locally free, the torsion part must vanish, i.e., we must have $\operatorname{Tor}(M) = 0$. In particular, this applies to locally free sheaves on the affine line $\mathbb{A}^1_k = \operatorname{Spec} k[x]$ over any field k.

Example 1.3 (The tangent bundle of the 2-sphere) Let $A = \mathbb{R}[x, y, z]/(x^2 + y^2 + z^2 - 1)$ and consider the A-linear map

$$\varphi : A^3 \longrightarrow A, \quad (a, b, c) \longmapsto ax + by + cz.$$

The kernel $M = \operatorname{Ker} \varphi$ defined a quasicoherent sheaf $\widetilde{M}$ which is locally free of rank 2.

To see this, note that M consists of triples $(a, b, c) \in A^3$ satisfying $ax + by + cz = 0$. After inverting x, any such triple is uniquely determined by b and c as $a = -x^{-1}(by + cz)$. This gives an isomorphism $M_x \cong A_x^2$, and hence $\widetilde{M}|_{D(x)} \cong \mathscr{O}^2_{D(x)}$. The same argument applies to y and z, so $\widetilde{M}$ is locally free.

While $\widetilde{M}$ is locally trivial, it is in fact globally non-trivial — M is not a free A-module. This is not so easy to prove: it reflects the famous topological fact that the tangent bundle of S^2 admits no nowhere-vanishing section (the Hairy Ball Theorem). In contrast, if we replace $\mathbb{R}$ by $\mathbb{C}$ in this example, the corresponding sheaf is trivial.

1.2 Transition functions

Let $\mathscr{E}$ be a locally free sheaf of rank r, and $\{U_i\}_{i\in I}$ be a trivializing open cover of $\mathscr{E}$. Then by definition, there are isomorphisms of $\mathscr{O}_{U_i}$-modules

$$\varphi_i : \mathscr{O}_{U_i}^r \xrightarrow{\sim} \mathscr{E}|_{U_i}. \tag{1.1}$$

Over the intersections $U_{ij} = U_i \cap U_j$ the maps $\tau_{ij} = \varphi_j^{-1} \circ \varphi_i$ are well-defined, and they give isomrophisms

$$\tau_{ij} : \mathscr{O}_{U_{ij}^r} \xrightarrow{\sim} \mathscr{O}_{U_{ij}}^r,$$

which restricted to the triple intersections $U_{ijk} = U_i \cap U_j \cap U_k$ satisfy the cocycle condition

$$\tau_{ik} = \tau_{jk} \circ \tau_{ij}. \tag{1.2}$$

Indeed, we have $\varphi_k^{-1} \circ \varphi_i = (\varphi_k^{-1} \circ \varphi_j) \circ (\varphi_j^{-1} \circ \varphi_i)$.

Conversely, we know from the Gluing Lemma for sheaves that given isomorphisms τ_{ij} as above, satisfying $\tau_{ji} = \tau_{ij}^{-1}$ and (1.2) on the triple overlaps, the sheaves $\mathscr{O}_{U_{ij}}^r$ may be glued together to a sheaf $\mathcal{E}$, which by definition is locally free of rank r.

Note that any isomorphism of $\mathscr{O}_{U_{ij}}$-modules $\mathscr{O}_{U_{ij}}^r \to \mathscr{O}_{U_{ij}}^r$ is given by some $r \times r$-matrix with entries in $\mathscr{O}_X(U_{ij})$. Thus it is possible to define $\mathscr{E}$ by specifying a collection of matrices τ_{ij} satisfying the cocycle condition (1.2). These are some times refered to as the **transition functions** of $\mathscr{E}$.

The isomorphisms τ_{ij} defining $\mathscr{E}$ are not unique. For instance, if we choose any isomorphisms $\psi_i : \mathscr{O}_{U_i}^r \to \mathscr{O}_{U_i}^r$, we can form new isomorphisms $\tau_{ij}' = \psi_j^{-1} \circ \tau_{ij} \circ \psi_i$, satisfying the cocycle conditions and which define the same sheaf $\mathscr{E}$. Furthermore, $\mathscr{E}$ could admit isomorphisms defined on a different open cover. This is therefore not an ideal way of classifying locally free sheaves, and in practice it is very difficult to decide when two constructions yield isomorphic sheaves.

Example 1.4 Suppose $\mathscr{E}$ and $\mathscr{F}$ are locally free of ranks r and s respectively. After refining, we may assume that they admit the same trivializing cover. Suppose that the gluing functions are given by τ_{ji} and ν_{ji} respectively. Then $\mathscr{E} \oplus \mathscr{F}$ is obtained by gluing together the different $\mathscr{O}_{U_i}^r \oplus \mathscr{O}_{U_i}^s$ with help of the matrices

$$\Phi_{ij} = \begin{pmatrix} \tau_{ij} & 0 \\ 0 & \nu_{ij} \end{pmatrix}.$$

1.3 Zero sets of sections

Definition 1.3.1 (Zero of sections)

*Let $\mathscr{E}$ be an locally free sheaf of rank r on the scheme X, and let $s \in \Gamma(X, \mathscr{E})$ be a global section. We define the **zero set of** s by*

$$V(s) := \{P \in X : s(P) = 0 \text{ in } \mathscr{E}(P)\}.$$

Equivalently, $V(s)$ consists of the points $P \in X$ such that $s_P \in \mathfrak{m}_P \mathscr{E}_P$.

The set $V(s)$ is a closed subset of X. In fact, it carries a canonical structure as a closed subscheme of X, which we will also denote by $V(s)$. To see this, consider an affine open subset $U = \operatorname{Spec} A$ such that $\mathscr{E}|_U \cong \mathscr{O}_X^r|_U$. Under this isomorphism, the restriction of s corresponds to an r-tuple of elements $(f_1, \dots, f_r) \in A^r$. This r-tuple generates an ideal $I = (f_1, \dots, f_r)$, which defines a closed subscheme $\operatorname{Spec} A/I \to \operatorname{Spec} A$. One can check that as U runs over all open affines where $\mathscr{E}$ is trivial, these closed subschemes glue together to form a closed subscheme of X.

The ideal sheaf describing $V(s)$ has the following description. If we identify $\Gamma(X, \mathscr{E}) = \mathrm{Hom}(\mathscr{O}_X, \mathscr{E})$, we may regard s as a map of $\mathscr{O}_X$-modules $s : \mathscr{O}_X \to \mathscr{E}$. Applying $\mathrm{Hom}_{\mathscr{O}_X}(\square, \mathscr{O}_X)$, we obtain a map of $\mathscr{O}_X$-modules

$$s^\vee : \mathscr{E}^\vee \longrightarrow \mathscr{O}_X. \tag{1.3}$$

The image of $s^\vee$ is a quasi-coherent ideal sheaf $\mathscr{I}$ of $\mathscr{O}_X$, and this is the ideal sheaf that defines $V(s)$. Concretely, over an affine open $U = \mathrm{Spec}\, A$, after identifying s with $(f_1, \dots, f_r) \in A^r$, the map $s^\vee$ is simply the tilde of the map $A^r \to A$ that sends the i-th basis vector e_i to f_i. In other words, $\mathscr{I}|_U$ corresponds to the ideal generated by the f_i's.

Proposition 1.3.1

Let X be an integral scheme and let $s \in \Gamma(X, \mathscr{E})$ be a global section of locally free sheaf $\mathscr{E}$ of rank r. The zero scheme $V(s)$ has codimension at most r in X.

In the special case when $\mathscr{E}$ has rank 1, then the map $s^\vee : \mathscr{E}^\vee \to \mathscr{O}_X$ is given by multiplication by s. In particular, if X is integral, this map is injective and identifies $\mathscr{E}^\vee$ with that ideal sheaf $\mathscr{I}$ of $V(s)$. The ideal sheaf sequence take the form

$$0 \longrightarrow \mathscr{E}^\vee \longrightarrow \mathscr{O}_X \longrightarrow \iota_* \mathscr{O}_{V(s)} \longrightarrow 0 \tag{1.4}$$

where $\iota : V(s) \to X$ is the inclusion.

1.4 Invertible sheaves and the Picard group

Among the locally free sheaves, the ones of rank 1 are particularly important. These are called **invertible sheaves**. By definition, an $\mathscr{O}_X$-module $\mathscr{L}$ is invertible whenever there exists an open cover $\{U_i\}_{i \in I}$ and isomorphism

$$\varphi_i : \mathscr{O}_{U_i} \xrightarrow{\sim} \mathscr{L}|_{U_i}.$$

These send the element $1 \in \mathscr{O}_X(U_i)$ to a section $s_i = \varphi_i(1) \in \mathscr{L}(U_i)$, which we call a **local generator** for $\mathscr{L}$ over U_i. For $i, j \in I$, both s_i and s_j generate $\mathscr{L}$ over $U_i \cap U_j$, so there is a relation

$$s_i = g_{ij} s_j$$

for some invertible element $g_{ij} \in \mathscr{O}_X(U_i \cap U_j)^\times$.

By condition, we have $g_{ii} = \mathrm{id}$ and $g_{ij} = g_{ji}^{-1}$. Moreover, over the triple overlaps $U_i \cap U_j \cap U_k$,

$$s_i = g_{ik} s_k, \quad s_i = g_{ij} s_j, \quad s_j = g_{jk} s_k,$$

which implies that the g_{ij} satisfy the **cocycle condition**

$$g_{ik} = g_{ij} \cdot g_{jk}$$

for all $i, j, k \in I$.

The invertible sheaf $\mathscr{L}$ is determined up to isomorphism by the collection of units $g_{ij} \in \mathscr{O}_X(U_{ij})^\times$. If $\mathscr{L}$ is trivia, i.e., there is an isomorphism $\mathscr{O}_X \to \mathscr{L}$, then the image of $1 \in \mathscr{O}_X(X)$ gives a section $s \in \mathscr{L}(X)$ that generates $\mathscr{L}$ everywhere. This means that there exists $g_i \in \mathscr{O}_X(U_i)^\times$ such that $s|_{U_i} = g_i s_i$ for every i. In terms of the g_{ij}, this means that $g_{ij} = g_j g_i^{-1}$.

As a special case of Proposition 1.1.2, we have:

Proposition 1.4.1

Let X be a scheme and $\mathscr{L}$ and $\mathscr{M}$ two invertible sheaves on X. Then:

(i) $\mathscr{L} \otimes_{\mathscr{O}_X} \mathscr{M}$ is also an invertible sheaf.

(ii) $\mathscr{H}om_{\mathscr{O}_X}(\mathscr{L}, \mathscr{M})$ is also invertible. In particular, $\mathscr{H}om_{\mathscr{O}_X}(\mathscr{L}, \mathscr{O}_X)$ is invertible, and

$$\mathscr{H}om_{\mathscr{O}_X}(\mathscr{L}, \mathscr{O}_X) \otimes_{\mathscr{O}_X} \mathscr{M} \xrightarrow{\sim} \mathscr{H}om_{\mathscr{O}_X}(\mathscr{L}, \mathscr{M}). \tag{1.5}$$

This implies that the tensor product acts as a sort of binary operation on the set of invertible sheaves; $\mathscr{L} \otimes \mathscr{M}$ is invertible if $\mathscr{L}$ and $\mathscr{M}$ are, and the tensor product is associative. Tensoring an invertible sheaf by $\mathscr{O}_X$ gives an isomorphism $\mathscr{L} \otimes_{\mathscr{O}_X} \mathscr{O}_X \cong \mathscr{L}$, so $\mathscr{O}_X$ serves as the identity. For an invertible sheaf $\mathscr{L}$, we define $\mathscr{L}^{-1} = \mathscr{H}om_{\mathscr{O}_X}(\mathscr{L}, \mathscr{O}_X)$. By the Proposition 1.4.1, we see that $\mathscr{L}^{-1}$ is again invertible, and serves as a multiplicative inverse of $\mathscr{I}$ under $\otimes$. In particular, this explains the term "invertible".

Definition 1.4.1 (Picard group)

For a scheme X, the ***Picard group*** $\operatorname{Pic}(X)$ *of X is the group of isomorphism classes of invertible sheaves on X under the tensor product.*

Note that it is the set of isomorphism classes of invertible sheaves that form a group, not the invertible sheaves themselves. That is, $\mathscr{L} \otimes_{\mathscr{O}_X} \mathscr{L}^{-1}$ is isomorphic, but strictly speaking, not equal to $\mathscr{O}_X$. Note also that $\operatorname{Pic}(X)$ is an abelian group, because $\mathscr{L} \otimes_{\mathscr{O}_X} \mathscr{M}$ is canonically isomorphic to $\mathscr{M} \otimes_{\mathscr{O}_X} \mathscr{L}$.

Example 1.5 (Spec $\mathbb{Z}$) Any invertible sheaf on $\operatorname{Spec}\mathbb{Z}$ is trivial (Example 1.1), so

$$\operatorname{Pic}(\operatorname{Spec}\mathbb{Z}) = 0.$$

On the other hand, $\operatorname{Pic}(\mathbb{Z}[\sqrt{-5}]) \neq 0$.

Example 1.6 (The affine line) Recall that every rank n vector bundle (locally free sheaf) on $\mathbb{A}_k^1$ is trivial of rank n, any invertible sheaf on $\mathbb{A}_k^1$ is trivial, so

$$\operatorname{Pic}(\mathbb{A}_k^1) = 0.$$

In fact, $\operatorname{Pic}(\mathbb{A}_k^n) = 0$ for any $n \geq 0$.

Example 1.7 Let $X \subseteq \mathbb{A}_k^2$ denote the plane curve defined by $y^2 = x^3 - x$ and let $P = (0,0)$. The ideal sheaf $\mathscr{I} \subseteq \mathscr{O}_X$ of sections vanishing at P is invertible. To see this, we show that for every $Q \in X$, the ideal $\mathscr{I}_Q$ is a principal ideal in $\mathscr{O}_{X,Q}$.

At the point P, the stalk $\mathscr{I}_P = (x,y)\mathscr{O}_{X,P}$. This ideal is generated by the element y. Indeed, since $y^2 = x(x-1)(x+1)$, and $(x-1)$ and $(x+1)$ are invertible in $\mathscr{O}_{X,P}$, we can express x in terms of y.

For any other point Q, at least x or y is invertible in $\mathscr{O}_{X,Q}$ and so $\mathscr{I}_Q = (x,y)\mathscr{O}_{X,Q} = (1)$ in $\mathscr{O}_{X,Q}$.

In either case, the ideal $\mathscr{I}_Q$ is principal, hence $\mathscr{I}$ is an invertible sheaf. It is however not trivial, as any ideal of sheaf of $\mathscr{O}_X$ is trivial if and only if it is principal, but (x,y) requires two generators.

On the other hand, $\mathscr{I} \otimes \mathscr{I}$ is trivial. Indeed, it is the sheaf associated to the ideal $(x,y)^2$ in $k[x,y]/(y^2 - x^3 - x)$, and

$$(x,y)^2 = (x^2, xy, y^2) = (x^2, xy, x^3 + x) = (x).$$

In terms of the Picard group, this means that the class $[\mathscr{I}]$ in $\operatorname{Pic}(X)$ is nonzero and satisfies $2[\mathscr{I}] = 0$. In fact, when k is algebraically closed, $\operatorname{Pic}(X)$ is an infinitely generated abelian group.

1.5 Pullbacks of invertible sheaves

Let $f : X \to Y$ be a morphism of schemes and $\mathscr{L}$ be an invertible sheaf on Y. Then $f^*\mathscr{L}$ is invertible on X. Indeed, if $\mathscr{L}$ is trivial over an open cover $\{V_i\}_{i\in I}$, then $f^*\mathscr{L}$ is trivial over the open sets $f^{-1}(V_i)$. Moreover, if $\mathscr{L}$ is described the local data $g_{ij} \in \mathscr{O}_Y(V_i \cap V_j)^\times$, then $f^*\mathscr{L}$ is described by the pullbacks $f^\sharp(g_{ij}) \in \mathscr{O}_X(f^{-1}(V_i \cap V_j))$.

If $\mathscr{L}$ and $\mathscr{M}$ are two invertible sheaves on Y, then

$$f^*(\mathscr{L} \otimes_{\mathscr{O}_Y} \mathscr{M}) = f^*(\mathscr{L}) \otimes_{\mathscr{O}_X} f^*(\mathscr{M}).$$

Therefore, the assignment $\mathscr{L} \mapsto f^*\mathscr{L}$ induces a morphism of groups

$$f^* : \mathrm{Pic}(Y) \longrightarrow \mathrm{Pic}(X).$$

The pullback operation extends naturally to sections. For any open subset $W \subseteq Y$ and section $s \in \mathscr{L}(W)$, we obtain a section $f^*s \in f^*\mathscr{L}(f^{-1}(W))$ as follows: locally on $W \cap V_i$ where $\mathscr{L}$ is trivial, we may write s as $a_i \cdot s_i$ for some $a_i \in \mathscr{O}_Y(W \cap V_i)$ and local generator s_i of $\mathscr{L}|_{V_i}$. The pullback section is then given by $f^*s = f^\sharp(a_i) \cdot f^*e_i$ on $f^{-1}(W \cap U_i)$, where f^*e_i is the corresponding local generators of $f^*\mathscr{L}|_{f^{-1}(V_i)}$. This construction is independent of the choice of trivialization and glues to a give a well-defined global section.

1.6 Important example: $\mathscr{O}_{\mathbb{P}^n_A}(m)$ sheaves

Projective space $\mathbb{P}^n_A$ carries a special collection of sheaves $\mathscr{O}_{\mathbb{P}^n_A}(m)$, where m is an integer. These are invertible sheaves, in the sense that they are locally isomorphic to the structure sheaf $\mathscr{O}_{\mathbb{P}^n_A}$.

The sheaves $\mathscr{O}_{\mathbb{P}^n_A}(m)$ are closely related to homogeneous polynomials. While homogeneous polynomials of degree $m \geq 1$ in $k[x_0, \dots, x_n]$ do not define regular functions on $\mathbb{P}^n_k$, they do define global sections of the sheaf $\mathscr{O}_{\mathbb{P}^n_k}(m)$. Intuitively, over an open set $U \subseteq \mathbb{P}^n_k$, the section of $\mathscr{O}_{\mathbb{P}^n_k}(m)$ can be identified with the set of quotients

$$\frac{f(x_0, \dots, x_n)}{g(x_0, \dots, x_n)} \quad \text{where } \deg f - \deg g = m$$

where f, g are homogeneous and $g(P) \neq 0$ for all $P \in U$.

Note *Denote* $x_{i/j} := \frac{x_i}{x_j}$.

To define these sheaves by gluing, we use the covering of $\mathbb{P}^n_A$ by $U_i = \operatorname{Spec} R_i$ where $R_i = A[x_{0/i}, \dots, x_{n/i}]$, as before. Over U_i, we define the R_i-module

$$M_i := A[x_{0/i}, \dots, x_{n/i}]x_i^m.$$

Note that since $x_i^m = x_{i/j}^m x_j^m$, there is an equality of A-modules

$$A[x_{0/i}, \dots, x_{n/i}, x_{i/j}]x_i^m = A[x_{0/j}, \dots, x_{n/j}, x_{j/i}]x_j^m. \tag{1.6}$$

In other word, we may identify the localizations

$$(M_i)_{x_{j/i}} = (M_j)_{x_{i/j}} \tag{1.7}$$

as modules over the ring $R_i[x_{i/j}] = R_j[x_{j/i}]$.

Note that M_i is a free R_i-module of rank 1, with x_i^m as a generator. Likewise, M_j is generated by x_j^m, and the elements in two localization are related by the formula

$$g(x_{0/i}, \dots, x_{n/i})x_i^m = x_{i/j}^m g\left(\frac{x_{0/j}}{x_{i/j}}, \dots, \frac{x_{n/j}}{x_{i/j}}\right) x_j^m. \tag{1.8}$$

By the equality (1.7), the two sheaves $\widetilde{M_i}$ and $\widetilde{M_j}$ on U_i restrict to the same sheaf on $U_{ij} = \operatorname{Spec}(R_i)_{x_{i/j}}$, namely the tilde of the module (1.6). As the isomorphisms are identity maps, the gluing conditions are satisfied, and the sheaves $\widetilde{M_i}$ over U_i glue to a sheaf on $\mathbb{P}^n_A$. We denote this sheaf by $\mathscr{O}_{\mathbb{P}^n_A}(m)$.

In the special case $m = 0$, we have $M_i = R_i$ and the sheaf $\mathscr{O}_{\mathbb{P}^n_A}(0) = \mathscr{O}_{\mathbb{P}^n_A}$ is simply the structure sheaf of $\mathbb{P}^n_A$.

Proposition 1.6.1

$\Gamma(\mathbb{P}^n_A, \mathscr{O}_{\mathbb{P}^n_A}(m))$ can be identified with free A-module of homogeneous polynomials of degree m in the variables $x_0, \ldots, x_n$, i.e., there is an isomorphism

$$\Gamma(\mathbb{P}^n_A, \mathscr{O}_{\mathbb{P}^n_A}(m)) \xrightarrow{\sim} A[x_0, \ldots, x_n]_m.$$

Proof By the sheaf sequence associated to the covering $U_0, \ldots, U_n$, a global section of $\mathscr{O}_{\mathbb{P}^n_A}(m)$ consists of section $s_i \in \Gamma(U_i, \mathscr{O}_{\mathbb{P}^n_A}(m))$ that agree on the over laps $U_i \cap U_j$. Over U_i we have $\Gamma(U_i, \mathscr{O}_{\mathbb{P}^n_A}(m)) = M_i$ and a section s_i is an element of the form

$$g_i\left(x_{0/i}, \ldots, x_{n/i}\right) x_i^m \in M_i. \tag{1.9}$$

The compatibility condition on overlaps implies that

$$g_j(x_{0/j}, \ldots, x_{n/j}) x_j^m = g_i(x_{0/i}, \ldots, x_{n/i}) x_i^m. \tag{1.10}$$

If $m < 0$, then the left hand side contains only terms with negative powers of x_j, whereas the right-hand side contains only negative powers of x_i. There is therefore only one tuple satisfying (1.10), namely where $g_0 = \cdots = g_n = 0$.

If $m \geq 0$, we see that the g_j are determined by g_0, i.e., $g_j(x_{0/j}, \ldots, x_{n/j}) = g_0(x_{1/0}, \ldots, x_{n/0}) x_{0/j}^m$. By comparing degrees in (1.10), we see that g_0 must have degree $\leq m$ as a polynomial in the $x_{1/0}, \ldots, x_{n/0}$. This means $F = g_0 x_0^m$ is a homogeneous polynomial of degree m in $x_0, \ldots, x_n$ with coefficients in A, and $s_i = g_i x_i^m = F$ for all $i = 0, \ldots, n$. Conversely, any homogeneous polynomial F of degree m determines a valid $(n+1)$-tuple $(s_0, \ldots, s_n)$. $\square$

The sheaves $\mathscr{O}_{\mathbb{P}^n_k}(m)$ highlight an important difference between $\mathbb{A}^n_k$ and $\mathbb{P}^n_k$. On $\mathbb{A}^n_k$, closed subsets are defined by the vanishing of regular functions — polynomials in $k[x_1, \ldots, x_n]$. In contrast, the only regular functions on $\mathbb{P}^n_k$ are the constants. However, we can define every closed subset of $\mathbb{P}^n_k$ as the zero set of some sections of the sheaves $\mathscr{O}_{\mathbb{P}^n_k}(m)$ (that is, homogeneous polynomials). For this section, invertible sheave play a central role in algebraic geometry.

Consider the projective line $\mathbb{P}^1_k$ over k, with the usual covering by $U_0 = \operatorname{Spec} k[u]$ and $U_1 = \operatorname{Spec} k[u^{-1}]$, glued together along their common open set $\operatorname{Spec} k[u, u^{-1}]$. The sheaves $\mathscr{O}_{\mathbb{P}^1_k}(n)$ are all quasi-coherent, because they are isomorphic to the structure sheaf $\mathscr{O}_{U_i}$ when restricted to the standard opens.

More generally, we can describe any quasicoherent sheaf on $\mathbb{P}^1_k$ as follows. A quasicoherent sheaf on $\mathbb{P}^1_k$ is given by a triple (M_0, M_1, τ), where M_0 is a module over $\mathscr{O}_X(U_0) = k[x_{1/0}]$ and M_1 is a module over $\mathscr{O}_X(U_1) = k[x_{0/1}]$ and

$$\tau : M_0 \otimes_{k[x_{1/0}]} k[x_{0/1}, x_{1/0}] \xrightarrow{\sim} M_1 \otimes_{k[x_{0/1}]} k[x_{0/1}, x_{1/0}]$$

is an isomorphism of modules over $k[x_{0/1}, x_{1/0}]$.

In terms of this description, the sheaves $\mathscr{O}_{\mathbb{P}^1_k}(n)$ are given by the triples $M_0 = k[x_{1/0}]x_0^n$ and $M_1 = k[x_{0/1}]x_1^n$ and

$$\begin{aligned} \tau : k[x_{0/1}, x_{1/0}]x_0^n &\xrightarrow{\sim} k[x_{0/1}, x_{1/0}]x_1^n \\ x_0^n &\longmapsto x_{0/1}^n \cdot x_1^n. \end{aligned}$$

Example 1.8 ($\mathscr{O}_{\mathbb{P}^1_k}(m)$ are invertible sheaves) On the projective line $X = \mathbb{P}^1_k$, the sheaves $\mathscr{O}_{\mathbb{P}^1_k}(m)$ are invertible sheaves, as they are constructed by gluing together two copies of $\mathscr{O}_{\mathbb{A}^1_k}$ over the standard covering of $\mathscr{O}_{\mathbb{A}^1_k}$ over the standard covering of $\mathbb{P}^1_k$. These sheaves are non-trivial for all $m \neq 0$, as we computed the global sections of $\mathscr{O}_{\mathbb{P}^1_k}(m)$ is not isomorphic to $\Gamma(\mathbb{P}^1_k, \mathscr{O}_{\mathbb{P}^1_k}) = k$.

The isomorphism $\mathscr{O}_{\mathbb{P}^1_A}(m) \otimes \mathscr{O}_{\mathbb{P}^1_A}(-m) \cong \mathscr{O}_{\mathbb{P}^1_A}$ shows that $\mathscr{O}_{\mathbb{P}^1_A}(-m)$ can be identified with the dual of $\mathscr{O}_{\mathbb{P}^1_A}(m)$.

Example 1.9 (Line bundles over $\mathbb{P}^1_k$) Every invertible sheaf $\mathscr{L}$ on $\mathbb{P}^1_k$ is isomorphic to $\mathscr{O}_{\mathbb{P}^1_k}(m)$ for some $m \in \mathbb{Z}$. This follows from the classification of quasicoherent sheaves on $\mathbb{P}^1_k$. Since any invertible sheaf on $\mathbb{A}^1_k$ is trivial, $\mathscr{L}$ is defined by $M_0 = k[x]$, $M_1 = k[x^{-1}]$ and an isomorphism

$$\tau : k[x, x^{-1}] \xrightarrow{\sim} k[x, x^{-1}].$$

However, any isomorphism $k[x, x^{-1}] \to k[x, x^{-1}]$ is obtained by multiplication by some x^m for $m \in \mathbb{Z}$. Therefore, $\mathscr{L}$ is defined by the same data as $\mathscr{O}_{\mathbb{P}^1_k}(m)$ and hence $\mathscr{L} \cong \mathscr{O}_{\mathbb{P}^1_k}(m)$.

Therefore, the map $m \mapsto [\mathscr{O}_{\mathbb{P}^1_k}(m)]$ defines an isomorphism

$$\mathrm{Pic}(\mathbb{P}^1_k) = \mathbb{Z}.$$

In fact, $\mathrm{Pic}(\mathbb{P}^n_k) = \mathbb{Z}$ for ant $n \geq 0$.

Example 1.10 (Pullback line bundles on $\mathbb{P}^2_k$) Consider the morphism

$$f : \mathbb{P}^1_k \longrightarrow \mathbb{P}^1_k; \quad [x_0 : x_1] \longmapsto [x_0^2 : x_0 x_1 : x_1^2].$$

In this example, $f^*\mathscr{O}_{\mathbb{P}^2_k}(1) = \mathscr{O}_{\mathbb{P}^2_k}(2)$. To see this, let U_0, U_1 and V_0, V_1, V_2 be the standard coverings of $\mathbb{P}^1_k$ and $\mathbb{P}^2_k$ respectively. Then the image of f is covered by the two open sets V_0 and V_2, and $f^{-1}V_0 = U_0$ and $f^{-1}V_2 = U_1$. We have $\mathscr{O}_{\mathbb{P}^2_k}(1)|_{V_0} y_0$ and $\mathscr{O}_{\mathbb{P}^2_k}(1)|_{V_2} = \mathscr{O}_{V_2} y_2$. The transition function between the two is given by $g_{02} = y_{2/0}$. Hence we have $f^\sharp(g_{02}) = (x_{1/0})^2$, so $f^*\mathscr{O}_{\mathbb{P}^2_k}(1)$ is defined by exactly the same transition functions as $\mathscr{O}_{\mathbb{P}^1_k}(2)$.

Note that the global sections of $\mathscr{O}_{\mathbb{P}^2_k}(1)$ are of the form $s = ay_0 + by_1 + cy_2$ for $a, b, c \in k$ (Proposition 1.6.1). These pullback to

$$f^*(s) = ax_0^2 + bx_0 x_1 + cx_2^2.$$

Example 1.11 Consider the closed subscheme $X = V(x_0) \subseteq \mathbb{P}^2_k$ and let $\iota : X \to \mathbb{P}^2_k$ be the inclusion. Then $\iota^*\mathscr{O}_{\mathbb{P}^2_k}(1) = \mathscr{O}_{\mathbb{P}^1_k}(1)$. Hence

$$\Gamma(\mathbb{P}^2_k, \mathscr{O}_{\mathbb{P}^2_k}(1)) \cong kx_0 \oplus kx_1 \oplus kx_2$$
$$\Gamma(X, \iota^*\mathscr{O}_{\mathbb{P}^2_k}(1)) \cong kx_1 \oplus kx_2,$$

by Proposition 1.6.1. In particular, contrary to f_*, the pullback functor does not always give an isomorphism on global sections.

Examples of pulling back sections

Example 1.12 Let $f : \mathbb{A}^2_k = \operatorname{Spec} k[x, y] \to \operatorname{Spec} k$ be the structure morphism. Then $f^{-1}\mathscr{O}_{\operatorname{Spec} k}$ is the constant sheaf on k on $\mathbb{A}^2_k$. The morphism $\mathscr{O}_{\operatorname{Spec} k} \to f_* f^{-1}\mathscr{O}_{\operatorname{Spec} k}$ is the identity map. On the other hand, $f_*\mathscr{O}_{\mathbb{A}^2_k}$ is the constant sheaf with value $k[x, y]$ on $\operatorname{Spec} k$ and $f^{-1}f_*\mathscr{O}_{\mathbb{A}^2_k}$ is the constant sheaf on $\mathbb{A}^2_k$ with value $k[x, y]$. The map $f^{-1}f_*\mathscr{O}_{\mathbb{A}^2_k} \to \mathscr{O}_{\mathbb{A}^2_k}$ is given by the inclusion $k[x, y] \to \mathscr{O}_{\mathbb{A}^2_k}(V)$ for each $V \subseteq \mathbb{A}^2_k$. Note that $f^{-1}f_*\mathscr{O}_{\mathbb{A}^2_k} \neq \mathscr{O}_{\mathbb{A}^2_k}$ in this example.

Example 1.13 For the structure morphism $f : \mathbb{P}^1_k \to \operatorname{Spec} k$ and $\mathscr{F} = \mathscr{O}_{\mathbb{P}^1_k}(-1)$, we have $f_*\mathscr{F} = \Gamma(\mathbb{P}^1_k, \mathscr{F})^\sim = \widetilde{(0)}$, and $f^{-1}f_*\mathscr{F} = 0$. In particular, $f^{-1}f_*\mathscr{F} \neq \mathscr{F}$.

As for the inverse image sheaf, we have canonical maps

$$\eta : \mathscr{G} \longrightarrow f_* f^* \mathscr{G} \tag{1.11}$$

and

$$f^* f_* \mathscr{F} \longrightarrow \mathscr{F}.$$

Here η is map of $\mathscr{O}_Y$-modules and ε is a map of $\mathscr{O}_X$-modules.

For a sections $s \in \mathscr{G}(V)$, we define the pullback to be the section

$$f^*(s) = \eta(s) \in \Gamma(f^{-1}V, f^*\mathscr{G}).$$

In the affine setting, these maps take a more concrete form:

Example 1.14 Let $f : \operatorname{Spec} B \to \operatorname{Spec} A$ be induced by a ring map $A \to B$, and $\mathscr{F} = \widetilde{M}$ and $\mathscr{G} = \widetilde{N}$. Then $\eta : \widetilde{N} \to f_* f^* \widetilde{N}$ corresponds to the map of A-modules

$$N \longrightarrow (N \otimes_A B)_A; \quad n \longmapsto (n \otimes 1).$$

Likewise, $\varepsilon : f^* f_* \widetilde{M} \to \widetilde{M}$ corresponds to the map of B-modules

$$M_A \otimes_A B \longrightarrow M; \quad m \otimes b \mapsto bm.$$

For $s \in \mathscr{G}(Y) = N$, the pullback $f^*(s) \in (f^*\mathscr{G})(X) = N \otimes_A B$ is given by $s \otimes 1$.

Example 1.14 is helpful for comuting pullbacks $f^*(s)$ for general morphisms: cover V and $f^{-1}(V)$ by affine schemes, then the affine pullbacks glue to a section $f^*(s)$.

Example 1.15 In the special case when $\mathscr{G} = \mathscr{O}_Y$, we have $f^*\mathscr{O}_Y = \mathscr{O}_X$ and the pullback of $s \in \mathscr{O}_Y(V)$ is simply given by

$$f^*(s) = f^\sharp(s) \in \mathscr{O}_X(f^{-1}V).$$

1.7 Morphisms to projective space

One reason to care about invertible sheaves is that they help us understanding morphisms to projective space. More precisely, given a scheme X, it is natural to ask what sort of data one needs to specify a morphism

$$f : X \longrightarrow \mathbb{P}^n. \tag{1.12}$$

The corresponding question for $\mathbb{A}^n$ has already been answered: morphisms $X \to \mathbb{A}^n$ are in one-to-one correspondence with elements of $\Gamma(X, \mathscr{O}_X)^n$, i.e., an n-tuple of regular functions on X.

For a morphism into projective space it will not be the global sections $\Gamma(X, \mathscr{O}_X)$, but rather sections of an invertible sheaf that will be the analogous data we need in order to specify a morphism.

To explain this properly, let $\mathscr{L}$ be an invertible sheaf and let $s \in \Gamma(X, \mathscr{L})$ be a global section. Consider the open set

$$D(s) = \{P \in X : s(P) \neq 0 \text{ in } \mathscr{L}(P)\}.$$

That is, $D(s)$ is the set of points $P \in X$ where $s_P \notin \mathfrak{m}_P \mathscr{L}_P$. Being the complement of $V(s)$, this set is indeed an open set of X.

Definition 1.7.1 (Globally generated invertible sheaf)

We say that ***a set of sections*** $\{s_i\}_{i \in I}$ ***generate*** $\mathscr{L}$ *if for any $P \in X$, there is an s_i such that $s_i(P) \neq 0$ in $\mathscr{L}(P)$. Equivalently, the open sets $D(s_i)$ form an open cover of X. If this holds, we say that $\mathscr{L}$ is* ***globally generated****.*

Lemma 1.7.1

Let X be a scheme and let $\mathscr{L}$ be an invertible sheaf on X. If $s \in \Gamma(X, \mathscr{L})$ is an nonzero global section, then there is an isomorphism of $\mathscr{O}_X$-modules

$$\varphi : \mathscr{O}_X|_{D(s)} \xrightarrow{\sim} \mathscr{L}|_{D(s)}$$

which sends 1 to s.

Proof We define φ over an open set $U \subseteq D(s)$, by sending $1 \in \mathscr{O}_X(U)$ to $s|_U \in \mathscr{L}(U)$ and extend $\mathscr{O}_X$-linearly. This is an isomorphism if and only if it is an isomorphism locally, so we may reduce to the case where $X = \operatorname{Spec} A$ and $\mathscr{L} = \mathscr{O}_X$. In that case, $D(s) = \operatorname{Spec} A_s$, s is invertible in A_s and so multiplication by s is an isomorphism $A_s \to A_s$. $\square$

Lemma 1.7.2

Let $f : X \to Y$ be a morphism of schemes and let $\mathscr{L}$ be an invertible sheaf on Y. Then if $\mathscr{L}$ is an invertible sheaf which is generated by global sections $s_0, \dots, s_n$, then $f^*\mathscr{L}$ is generated by the sections $t_0 = f^*s_0, \dots, t_n = f^*s_n$, and X is covered by the open sets $D(t_0), \dots, D(t_n)$.

Proof As $\mathscr{L}$ is invertible, we may reduce to the case $X = \operatorname{Spec} B$, $Y = \operatorname{Spec} A$, $\mathscr{L} = \mathscr{O}_Y$ and f is induced by a ring map $\varphi : A \to B$. Then we can view the sections $s_0, \dots, s_n$ as elements of A, and the pullback $f^*(s)$ of a section s is simply given by $\varphi(s)$.

Let $P \in X$ be a point. Then there is an index j such that $s_j(Q) \neq 0 \in \kappa(Q)$ is nonzero at $Q = f(P)$. Then $f^*(s_j) = \varphi(s_j)$ is then also nonzero at P, as its class in $\mathscr{O}_{X,P}$ coincides with the image of s_j via the map of local rings $f_P : \mathscr{O}_{Y,P} \to \mathscr{O}_{X,P}$.

As the elements $s_0, \dots, s_n$ generate $\mathscr{L}$, they generate the unit ideal in A, and so the pullbacks $\varphi(s_0), \dots, \varphi(s_n)$ generate the unit ideal in B. Hence $X = \operatorname{Spec} B$ is covered by the open sets $D(f^*(s_0)), \dots, D(f^*(s_n))$. $\square$

Example 1.16 If $X = \operatorname{Spec} A$ is an affine scheme, then any invertible sheaf $\mathscr{L}$ is globally generated. Since $\mathscr{L}$ is quasicoherent sheaf, $\mathscr{L} = \widetilde{M}$ for some A-module M. Choosing a (possibly infinite) set of generators $\{m_i\}_{i \in I}$ for M, these define global sections of $\mathscr{L}$ which generate $\mathscr{L}$ in every stalk.

Example 1.17 On $\mathbb{P}^n_k$, the sections $x_0, \dots, x_n$ generate $\mathscr{O}_{\mathbb{P}^n_k}(1)$. Indeed, every point $P \in \mathbb{P}^n_k$ lies in some open set $U_i \subseteq \mathbb{P}^n_k$, and for this i, we have $x_i(P) \neq 0$.

Combining the Example 1.17 with Lemma 1.7.2, we see that given a morphism $f : X \to \mathbb{P}^n$, we obtain an invertible sheaf $\mathscr{L} = f^*\mathscr{O}_{\mathbb{P}^n}(1)$ on X, as well as $n+1$ distinguished global sections $s_i = f^*x_i$ by pulling back the sections $x_0, \dots, x_n$ of $\mathscr{O}(1)$. As $\mathscr{O}(1)$ is globally generated by $x_0, \dots, x_n$, $\mathscr{L}$ will be globally generated by the pullbacks $s_0, \dots, s_n$.

The main result in this section is that there is a way to reverse this process. In other words, from a given invertible sheaf $\mathscr{L}$ and $n+1$ global sections $s_i \in \Gamma(X, \mathscr{L})$ that generate $\mathscr{L}$ at every point, we can uniquely construct a morphism $f : X \to \mathbb{P}^n$ such that $f^*\mathscr{O}_{\mathbb{P}^n}(1) = \mathscr{L}$ and $f^*x_i = s_i$. Hence $(L, s_0, \dots, s_n)$ is the data we are after.

Theorem 1.7.1

Let X be a scheme over a ring A, and let $\mathscr{L}$ be an invertible sheaf on X with global sections $s_0, \dots, s_n \in \Gamma(X, \mathscr{L})$ that generate $\mathscr{L}$. Then there is a unique morphism of A-schemes

$$f : X \longrightarrow \mathbb{P}^n_A = \operatorname{Proj} A[x_0, \dots, x_n]$$

and an isomorphism $f^*\mathscr{O}(1) \cong \mathscr{L}$ which maps f^*x_i to s_i for $i = 0, \dots, n$.

Proof Write $\mathbb{P}^n_A = \operatorname{Proj} R$, where $R = A[x_0, \dots, x_n]$ and denote $\mathscr{O}_{\mathbb{P}^1_A}(1)$ and denote $\mathscr{O}_{\mathbb{P}^1_A}(1)$ by $\mathscr{O}(1)$.

By assumption, the $n+1$ sections $s_0, \dots, s_n$ generate invertible sheaf $\mathscr{L}$. By Lemma 1.7.1, the open sets $D(s_i)$ provide a local trivializing cover of $\mathscr{L}$, with isomorphisms $\varphi_i : \mathscr{O}_X|_{D(s_i)} \to \mathscr{L}|_{D(s_i)}$ which sends 1 to the section s_i. Since s_i is a local generator for $\mathscr{L}$ over $D(s_i)$ we may for each $j = 0, \dots, n$, express $s_j|_{D(s_i)}$ as $r_{ij}s_i$ for some $r_{ij} \in \mathscr{O}_X(D(s_i))$. For simplicity, we write $s_{j/i}$ for r_{ij}.

These section define a map of A-algebras

$$(R_{x_i})_0 \longrightarrow \Gamma(D(s_i), \mathscr{O}_X); \quad x_{j/i} \longmapsto s_{j/i} \tag{1.13}$$

By the correspondence between ring homomorphisms and maps into affine schemes, we obtain a morphism of A-schemes $f_i : D(s_i) \to D_+(x_i)$.

To check that the morphisms f_i glue to a morphism $f : X \to \mathbb{P}^n_A$, we check the overlaps $D(s_i) \cap D(s_j)$ and $D_+(x_i) \cap D_+(x_j) = D_+(x_i x_j)$. Note that there is a commutative diagram.

$$\begin{tikzcd}
(R_{x_i})_0 \arrow[r] \arrow[d] & \Gamma(D(s_i), \mathscr{O}_X) \arrow[d] \\
(R_{x_i x_j})_0 \arrow[r] & \Gamma(D(s_i s_j), \mathscr{O}_X)
\end{tikzcd} \tag{1.14}$$

As the bottom the map sends $x_{k/i} = \frac{x_{k/j}}{x_{i/j}}$ to $s_{k/i} = \frac{s_{k/j}}{s_{i/j}}$, we see that both f_i and f_j restrict to the same map $D(s_i s_j) \to \mathbb{P}^n_A$, namely the one induced by the bottom map in (1.14). Hence they glue to a morphism $f : X \to \mathbb{P}^n$.

Next, we show that there is an isomorphism $f^*\mathscr{O}(1) \cong \mathscr{L}$. On $\mathbb{P}^n_A$, we have $\mathscr{O}(1)|_{D_+(x_i)} = \mathscr{O}_{D_+(x_i)} x_i$, and x_i serves as the local generator. Over the open sets $D(s_i) = f^{-1}(D(x_i))$, we therefore have $f^*\mathscr{O}(1)|_{D(s_i)} \cong \mathscr{O}_{D(s_i)}$ with local generator f^*x_i. Likewise, $\mathscr{L}|_{D(s_i)} \cong \mathscr{O}_{D(s_i)}$ with local generator s_i. As both $\mathscr{L}$ and $f^*\mathscr{O}(1)$ are trivial over $D(s_i)$, we may define an isomorphism $f^*\mathscr{O}(1)|_{D(s_i)} \to \mathscr{L}|_{D(s_i)}$ by sending f^*x_i to s_i. To check that these isomorphisms glue, we need to check that they are compatible with the gluing maps. The sheaf $\mathscr{O}(1)$ is obtained by gluing together $\mathscr{O}_{D(x_i)}$ using the transition functions $g_{ij} = x_{j/i}$. Therefore $f^*\mathscr{O}(1)$ is obtained by gluing together the $\mathscr{O}_{D(s_i)}$ using multiplication by $f^*g_{ij} = f^\sharp(x_{j/i})$ as transition functions. However, by construction, $f^\sharp(x_{j/i}) = s_{j/i}$. Hence $f^*\mathscr{O}(1)$ and $\mathscr{L}$ are defined by the same gluing data, and the local isomorphisms define an isomorphism $f^*\mathscr{O}(1) \cong \mathscr{L}$. It is clear that the generator f^*x_i maps to s_i via this identification, because this happens over each open set $D(s_i)$.

The uniqueness part is clear, because the sections $s_0, \dots, s_n$ determine the ratios $s_{j/i}$, which inturn determine f. □

We will refer to a morphism $\varphi : X \to \mathbb{P}^n_A$ as given by the data $(\mathscr{L}, s_0, \dots, s_n)$ and informally write

$$X \longrightarrow \mathbb{P}^n_A x \quad \longmapsto [s_0(x) : \cdots : s_n(x)]$$

One should still keep in mind that the sections s_i are sections of $\mathscr{L}$, not regular functions. In light of the above proof, we see that it is the ratios $s_{j/i}$ which can be interpreted as regular functions, locally on $D(s_i) = \{x \in X : s_i(x) \neq 0\}$.

We also see that two sets of data $(\mathscr{L}, s_0, \dots, s_n)$, $(\mathscr{L}, t_0, \dots, t_n)$ give rise to the same morphism $f : X \to \mathbb{P}^n_A$ if and only if there is a unit $\lambda \in \mathscr{O}_X^\times(X)$ such that $t_i = \lambda s_i$ for each i. Thus morphisms $f : X \to \mathbb{P}^n_A$ are in bijective correspondence with the data $(\mathscr{L}, s_0, \dots, s_n)$ modulo this equivalence relation. One can compare this with the description of the A-valued points of $\mathbb{P}^n_A$.

Example 1.18 Let $X = \operatorname{Spec} \mathbb{Z}$. Since any invertible sheaf on $\operatorname{Spec} \mathbb{Z}$ is isomorphic to the structure sheaf, we see that $\mathbb{Z}$-morphisms $f : X \to \mathbb{P}^n_{\mathbb{Z}}$ are in one-to-one correspondence with $(n+1)$-tuples of elements in $\Gamma(X, \mathscr{O}_X) = \mathbb{Z}$. In other words, and $\mathbb{Z}$-point $\operatorname{Spec} \mathbb{Z} \to \mathbb{P}^n_{\mathbb{Z}}$ is in "homogeneous coordinate form". The same applies when A is a local ring, or any ring with $\operatorname{Pic}(\operatorname{Spec} A) = 0$. On the other hand, $\operatorname{Pic}(\operatorname{Spec}\mathbb{Z}[\sqrt{-5}]) \neq 0$, and we say that there were "non-obvious" maps $\operatorname{Spec}\mathbb{Z}[\sqrt{-5}] \to \mathbb{P}^1_{\mathbb{Z}}$.

Example 1.19 Let $X = \mathbb{P}^1_k = \operatorname{Proj} k[s,t]$ and $\mathscr{L} = \mathscr{O}_{\mathbb{P}^1_k}(2)$. Then $\mathscr{L}$ is globally generated by s^2, st, t^2 and the corresponding morphism

$$\begin{aligned} \varphi : \mathbb{P}^1_k &\longrightarrow \mathbb{P}^2_k \\ (s:t) &\longmapsto (s^2 : st : t^2) \end{aligned}$$

has image $V(x_0x_2 - x_1^2)$ which is an irreducible degree 2 curve.

Example 1.20 (Cuspidal cubic) Let $X = \mathbb{A}^1_k$ and $\mathscr{L} = \mathscr{O}_X$. Then $\Gamma(X, \mathscr{L}) = k[t]$. Choosing the three sections $1, t, t^2$, we get a map of schemes

$$\begin{aligned} \varphi : X &\longrightarrow \mathbb{P}^2_k \\ t &\longrightarrow [1 : t^2 : t^3] \end{aligned}$$

whose image in $\mathbb{P}^2$ is the cuspidal cubic $V(x_0x_2^2 - x_1^3)$ minus the point at infinity.

Remark **Important Remark:** Given a scheme X with $s_0, \ldots, s_n$ of an invertible sheaf $\mathscr{L}$, there is a maximal open subset U such that the sections generate $\mathscr{L}$ for all points in U, namely $U = \bigcup_{i=0}^{n} D(s_i)$. Even if we do not assume that the s_i globally generate $\mathscr{L}$, we still get a morphism $\varphi : U \to \mathbb{P}^n_A$. In other words, the sections define a rational map $\varphi : X \dashrightarrow \mathbb{P}^n_A$, which is a morphism when restricted to U.

Example 1.21 Let $X = \mathbb{A}^{n+1}_k = \operatorname{Spec} k[x_0, \ldots, x_n]$ and $\mathscr{L} = \mathscr{O}_X$. Then $\Gamma(X, \mathscr{L}) = k[x_0, \ldots, x_n]$. If we take the sections $x_0, \ldots, x_n$, then they generate L outside $V(x_0, \ldots, x_n)$. Hence we get a morphism of schemes

$$\begin{aligned} \mathbb{A}^{n+1}_k \setminus V(x_0, \ldots, x_n) &\longrightarrow \mathbb{P}^n_k \\ (x_0, \ldots, x_n) &\longmapsto [x_0 : \cdots : x_n] \end{aligned}$$

which is exactly the "quotient space" description of $\mathbb{P}^n$.

Chapter 2 Sheaves on projective space

In this chapter we study quasi-coherent sheaves on projective schemes. As in the affine case, there is a functor $\widetilde{(-)}$ that assigns to each graded R-module M a quasi-coherent sheaf $\widetilde{M}$ on $\operatorname{Proj} R$. The main result is that every quasi-coherent sheaf on a projective scheme arises in this way. In particular, this framework allows us to describe closed subschemes of a projective scheme: every closed subscheme is determined by a homogeneous ideal.

Contrary to the affine case, however, the functor $\widetilde{(-)}$ is not an equivalence of categories: different graded modules can determine the same quasi-coherent sheaf, and distinct homogeneous ideals may define the same closed subscheme. This is one aspect in which projective schemes are more subtle than affine schemes (another being that morphisms of affine schemes are described in terms of ring maps, whereas morphisms or projective schemes are in general not).

2.1 Quasicoherent sheaves on projective schemes

Let R be a graded ring. If M is a graded R-module, we define a sheaf $\widetilde{M}$ by setting, for each $D_+(f) \subseteq \operatorname{Proj} R$, where f is homogeneous of positive degree,

$$\Gamma(D_+(f), \widetilde{M}) = (M_f)_0. \tag{2.1}$$

For an inclusion $D_+(fg) \subseteq D_+(f)$, there is a canonical localization map of graded modules $(M_f)_0 \to (M_{fg})_0$. Taking degree 0 parts, we have a canonical map

$$(M_f)_0 \longrightarrow (M_{fg})_0. \tag{2.2}$$

This defines a restriction map $\Gamma(D_+(f), \widetilde{M}) \to \Gamma(D_+(fg), \widetilde{M})$. The $\mathscr{O}_X$-module axioms are easily verified.

Proposition 2.1.1

The sheaf $\widetilde{M}$ is a quasicoherent sheaf on $\operatorname{Proj} R$.

Note that a map of graded modules $M \to N$ induces maps between the degree 0 part of the localizations $(M_f)_0 \to (N_f)_0$. These are compatible with further localizations, so we get induced maps of $\mathscr{O}_X$-modules $\widetilde{M} \to \widetilde{N}$. It follows that the graded tilde operation $M \mapsto \widetilde{M}$ defines a functor from the category of graded R-modules to quasi-coherent sheaves on X.

Each element $m \in M_0$ determines a section of $\Gamma\left(D_+(f), \widetilde{M}\right) = (M_f)_0$, by taking the image of m in $(M_f)_0$. If g is another homogenous element, then the two sections clearly agree when restricted to the overlaps $D_+(fg)$, (they both map to the image of m in $(M_{fg})_0$). Therefore, they glue together to a global section of $\widetilde{M}$.

The following proposition summarizes the basic properties of the tilde-functor.

Proposition 2.1.2

Let R be a graded ring. The graded tilde functor $M \mapsto \widetilde{M}$ has the following properties:

(i) It is addictive and exact.

(ii) For each $\mathfrak{p} \in \operatorname{Proj} R$, we have $(\widetilde{M})_{\mathfrak{p}} = (M_{\mathfrak{p}})_0$.

(iii) If M is finitely generated, then $\widetilde{M}$ is of finite type.

From now on, we will let $X \subseteq \mathbb{P}^n_A$ be a projective scheme over A. That is $X = \operatorname{Proj} R$, where R is a graded ring of the form

$$R = A[x_0, \ldots, x_n]/\mathfrak{a} \tag{2.3}$$

where x_i have degree 1 and $\mathfrak{a} \subseteq A[x_0, \ldots, x_n]$ is a homogeneous ideal. (This assumption is not too restrictive.)

In many aspects the projective tilde-functor behaves as the one of Spec, but there are also many important differences. The most striking is that different non-isomorphic modules can give rise to the same sheaf. This can be explained from the fact that primes contained in $V(R_+)$ are disregarded in the Proj-construction — this has the effect that graded modules supported in $V(R_+)$ must give the zero sheaf after applying the tilde-functor.

Note *For any integer d, we let $M_{>d}$ be the R-module $M_{>d} = \bigoplus_{i>d} M_i$.*

Lemma 2.1.1

Let M and N be two graded R-modules,

(i) $\widetilde{M} = 0$ if and only if $\operatorname{Supp} M \subseteq V(x_0, \ldots, x_n)$.

(ii) Assume that $M_{>d} \cong N_{>d}$ for some d. Then $\widetilde{M} \cong \widetilde{N}$. The converse also holds if M and N are finitely generated.

Proof To prove (i), suppose that $\operatorname{Supp} M \subseteq V(R_+)$, then Proposition 2.1.2 implies that the stalks of $\widetilde{M}$ are zero for every $\mathfrak{p} \in \operatorname{Proj} R$, and hence $\widetilde{M} = 0$. Conversely, if the support of M is not contained in $V(R_+)$, there is a homogeneous prime ideal $\mathfrak{p} \in \operatorname{Proj} R$ such that $M_\mathfrak{p} \neq 0$. Suppose R is generated in degree 1, we want to show that $(M_\mathfrak{p})_0 \neq 0$. Indeed, $\operatorname{Proj} R$ is covered by distinguished open sets $D_+(f)$ where f has degree 1, and so there is an f of degree 1 not lying in $\mathfrak{p}$. Then for an nonzero homogeneous element $x \in M_\mathfrak{p}$, the element $\frac{x}{f^{\deg x}}$ yields a nonzero element of degree 0 in $M_\mathfrak{p}$, so $(M_\mathfrak{p})_0 \neq 0$. This contradicts to the fact that $\widetilde{M} = 0$.

To prove (ii), note that the quotient $K = M/M_{>d}$ is a graded module which is killed by the power $(R_+)^d$ and consequently has support in $V(R_+^d) = V(R_+)$. By (i), we have $\widetilde{K} = 0$, and hence $\widetilde{M_{>d}} = \widetilde{M}$. As this holds for N as well, we get $\widetilde{M} \cong \widetilde{M_{>d}} \cong \widetilde{N_{>d}} \cong \widetilde{N}$. $\square$

Example 2.1 On $\mathbb{P}^1_k = \operatorname{Proj} k[x_0, x_1]$, the module $M = k[x_0, x_1]/(x_0^2, x_1^2)$ satisfies $\widetilde{M} = 0$, but it is nonzero.

Tensor products

Let M and N be two graded R-modules. There is a natural way of defining a grading on the tensor product $M \otimes_R N$, by defining the degree d piece $(M \otimes_R N)_d$ to be the additive subgroup of $M \otimes_R N$ generated by elements $x \otimes y$ where $x \in M_p$ and $y \in N_q$ where $p + q = d$. One checks that $M \otimes_R N$ is the direct sum of these graded parts (as an R_0-module).

Example 2.2 Let $R = k[x]$ with the standard grading, and let $M = N = R$. Then $(M \otimes_R N)_d = k[x]_d$. Note that the graded pieces of $(M \otimes_R N)_d$ are not equal to $\bigoplus_{p+q=d}(M_p \otimes_{R_0} N_q)$. In fact, $\bigoplus_{p+q=d}(M_p \otimes_{R_0} N_q) \cong k[x, y]_d$.

Proposition 2.1.3

For graded R-modules M and N, there is a natural isomorphism

$$\widetilde{M} \otimes_{\mathscr{O}_{\operatorname{Proj} R}} \widetilde{N} \xrightarrow{\sim} \widetilde{M \otimes_R N}.$$

Proof For each homogeneous element $f \in R$ of positive degree, there is a canonical map

$$M_f \otimes_{(R_f)_0} N_f \longrightarrow M_f \otimes_{R_f} N_f \cong (M \otimes_R N)_f$$

induced by sending $\frac{x}{f^n} \otimes \frac{y}{f^m}$ to $\frac{x \otimes y}{f^{n+m}}$. When restricted to elements of degree 0, we get a map

$$(M_f)_0 \otimes_{(R_f)_0} (N_f)_0 \longrightarrow ((M \otimes_R N)_f)_0. \tag{2.4}$$

It is easy to check this is an $\mathscr{O}_{\operatorname{Proj} R}$-module morphism. Therefore, we get a natural map of sheaves

$$\widetilde{M} \otimes_{\mathscr{O}_{\operatorname{Proj} R}} \widetilde{N} \longrightarrow \widetilde{M \otimes_R N}. \tag{2.5}$$

This is an isomorphism, because it is an isomorphism over each f of degree 1. □

2.2 The sheaves $\mathscr{O}(m)$

The most important sheaves on $\operatorname{Proj} R$ are the so-called **twisting sheaves**, denoted by $\mathscr{O}_{\operatorname{Proj} R}(d)$. These generalize the sheaves $\mathscr{O}_{\mathbb{P}^n_A}(d)$ on $\mathbb{P}^n_A$ which were introduced in Section 1.6. Just as homogeneous polynomials in $A[x_0, \dots, x_n]_d$ define global sections of $\mathscr{O}_{\mathbb{P}^n_A}(d)$, the degree d elements of R will define sections of the sheaf $\mathscr{O}_{\operatorname{Proj} R}(d)$.

If M is a graded R-module, and n is an integer, we define an R-module $M(n)$ as follows: the underlying R-module of $M(n)$ is just M, but the grading is shifted by d steps:

$$M(n)_d = M_{d+n}. \tag{2.6}$$

That is, $N = M(n)$ is a graded R-module such that $N_0 = M_n$, $N_1 = M_{n+1}$ and so on, and elements from M_d considered as elements in N will be of degree $d - n$.

In the special case when $M = R$, this gives a graded and free R-module $R(n)$, which is generated by the element $1 \in R_{-n}$. Note the canonical isomorphism $M(n) = M \otimes_R R(n)$: both have M as underlying module, and the gradings coincide by the way we have defined the grading on the tensor product. Moreover, for each $m, n \in \mathbb{Z}$, we have

$$R(n) \otimes_R R(m) = R(n+m). \tag{2.7}$$

Applying the tilde-functor to $R(n)$ gives us a quasicoherent $\mathscr{O}_{\operatorname{Proj} R}$-module on $\operatorname{Proj} R$:

Definition 2.2.1 (Twisting sheaf)

For each integer $n \in \mathbb{Z}$, and for $X = \operatorname{Proj} R$, we define

$$\mathscr{O}_X(n) = \widetilde{R(n)}.$$

For an $\mathscr{O}_X$-module $\mathscr{F}$, we define the twist of $\mathscr{F}$ by n to be the $\mathscr{O}_X$-module

$$\mathscr{F}(n) = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{O}_X(n).$$

If $h \in R$ is homogeneous of degree d, then it defines a section of $\mathscr{O}_X(d)$ via $\Gamma(X, \mathscr{O}_X(d)) \cong R_d$. Abusing notation slightly, we will continue to write h for this section.

For an $\mathscr{O}_X$-module $\mathscr{F}$, the tensor product gives us natural multiplication maps

$$\mathscr{O}_X(d) \otimes_{\mathscr{O}_X} \mathscr{F}(e) \longrightarrow \mathscr{F}(d+e).$$

In the case $\mathscr{F} = \widetilde{M}$, this is simply the map of sheaves induced by the multiplication maps

$$\begin{aligned} (R_f)_d \otimes_{(R_f)_0} (M_f)_e &\longrightarrow (M_f)_{d+e} \\ \frac{h}{f^a} \otimes \frac{m}{f^b} &\longmapsto \frac{h \cdot m}{f^{a+b}} \end{aligned}$$

over each distinguished open.

The sheaf $\mathscr{O}_X(m)$ is an invertible sheaf. As x_i is invertible in R_{x_i}, we have $R_{x_i} \cdot x_i^m \cong R_{x_i}$. In degree 0, we find that

$$(R(m)_{x_i})_0 = (R_{x_i})_m = x_i^m \cdot (R_{x_i})_0.$$

Therefore, on the distinguished affine open set $D_+(x_i)$, we see that

$$\mathscr{O}_X(m)|_{D_+(x_i)} = x_i^m \cdot \mathscr{O}_X|_{D_+(x_i)}.$$

Moreover, for each $m, n \in \mathbb{Z}$, there are canonical isomorphisms

$$\mathscr{O}_X(m+n) \cong \mathscr{O}_X(m) \otimes_{\mathscr{O}_X} \mathscr{O}_X(n).$$

For graded rings with non-standard gradings, one can still define $\widetilde{M}$ and $\mathscr{O}_X(m)$ but some of the results are more complicated. For instance, it can happen that $\widetilde{M} \otimes \widetilde{N} \not\cong \widetilde{M \otimes_R N}$ and $\mathscr{O}(m)$ is not be locally free. Here is an example:

Example 2.3 Consider the polynomial ring $R = k[x_0, x_1, x_2]$ with a grading defined by $\deg x_0 = 1, \deg x_1 = 2, \deg x_2 = 3$. After localizing at x_2, the degree 0 part of R_{x_2} is given by

$$(R_{x_2})_0 = k\left[\frac{x_0^3}{x_2}, \frac{x_0 x_1}{x_2}, \frac{x_1^3}{x_2^2}\right].$$

We claim that $\widetilde{M} \otimes \widetilde{N} \neq \widetilde{M \otimes N}$ for the two graded R-modules $M = R(1)$ and $N = R(1)$. Of cause, we have $R(1) \otimes R(1) = R(2)$. However, we have the following equalities of $(R_{x_2})_0$-modules:

$$(R(1)_{x_2})_0 = (R_{x_2})_0 \cdot x_0 + (R_{x_2})_0 \cdot \frac{x_1^2}{x_2}$$
$$(R(2)_{x_2})_0 = (R_{x_2})_0 \cdot x_1 + (R_{x_2})_0 \cdot x_0^2.$$

Next, if we compute the tensor product of $(R(1)_{x_2})_0$ with itself over $(R_{x_2})_0$ we get an $(R_{x_2})_0$-module which does not contain the monomial x_1. However, x_1 clearly belongs to $(R(2)_{x_2})_0$. This shows that $\widetilde{R(1)} \otimes \widetilde{R(1)} \not\cong \widetilde{R(2)}$, as the two sheaves are different over the open set $U = D_+(x_2)$.

2.3 Sheaves on projective schemes

Here is the main theorem of this section:

Theorem 2.3.1

Every quasicoherent sheaf $\mathscr{F}$ on projective scheme X is of the form $\widetilde{M}$ for some graded R-module M. If $\mathscr{F}$ is of finite type, then $\mathscr{F} \cong \widetilde{M}$ for some finitely generated graded R-module.

In the course of the proof, we will see that we may take M to be the module

$$\Gamma_*(\mathscr{F}) = \bigoplus_{d \in \mathbb{Z}} \Gamma(X, \mathscr{F}(d)).$$

This is called the **associated graded module** to $\mathscr{F}$. The R-module structure is defined via the multiplication maps

$$\mathscr{O}_X(d) \otimes \mathscr{F}(e) \longrightarrow \mathscr{F}(d+e).$$

More precisely, if $a \in R_d$, and $s \in \Gamma_*(\mathscr{F})_e$, we regard a as a section of $\mathscr{O}_X(d)$ and define the product $h \cdot s$ to be the image of $h \otimes s$ in $\Gamma(X, \mathscr{F}(d+e)) = \Gamma_*(\mathscr{F})_{d+e}$.

The module $\Gamma_*(\mathscr{F})$ plays the role of the global sections functor on $\operatorname{Spec} A$. For projective schemes, it is not enough to consider the global sections alone; one has to consider the global sections of all twists of $\mathscr{F}$ at once, to be able to recover $\mathscr{F}$.

Lemma 2.3.1 (Quasicoherence of pushforward)

Let $f : X \to Y$ be a morphism and suppose that there exists an affine open cover $\{V_i\}_{i\in I}$ of Y such that for every $V \in \{V_i\}$, the preimage $f^{-1}(V)$ is a finite union of affine open sets $U_1, \ldots, U_r$ and each intersection $U_i \cap U_j$ is also a finite union of affine open sets. Then for any quasicoherent $\mathscr{O}_X$-module $\mathscr{F}$, the pushforward $f_*\mathscr{F}$ is a quasicoherent on Y.

We now proof Theorem 2.3.1

Proof Say $X = \operatorname{Proj} R$, where $R = k[x_0, \ldots, x_n]/\mathfrak{a}$. Let $U = \operatorname{Spec} R \setminus V(x_0, \ldots, x_n)$, and consider the two morphisms $\iota : U \to \operatorname{Spec} R$ and $\pi : U \to X$ where $\pi((a_0, \ldots, a_n)) = [a_0 : \cdots : a_n]$. These fit into the following diagram:

$$\begin{tikzcd} U \arrow[r, hook, "\iota"] \arrow[d, "\pi"] & \operatorname{Spec} R \\ X \end{tikzcd}$$

Since $\mathscr{F}$ is quasicoherent on X, the pullback $\pi^*\mathscr{F}$ is quasicoherent on U. The morphism ι satisfies the conditions of Lemma 2.3.1, so the pushforward $\iota_*\pi^*\mathscr{F}$ is quasicoherent on $\operatorname{Spec} R$. Therefore,

$$\iota_*\pi^*\mathscr{F} = \widetilde{M}$$

for some R-module M. We can identify this module as

$$M = \Gamma(\operatorname{Spec} R, \iota_*\pi^*\mathscr{F}) = \Gamma(U, \pi^*\mathscr{F}).$$

We will relate M to the associated graded module using the following lemma:

Lemma

There is a natural isomorphism of $\mathscr{O}_X$-modules $\pi_*\pi^*\mathscr{F} = \bigoplus_{d\in\mathbb{Z}} \mathscr{F}(d)$, where X is a projective scheme.

Proof We use the standard affine covering of X by the distinguished open sets $D_+(x_i)$. For each i, the preimage $\pi^{-1}(D_+(x_i)))$ is the distinguished open set $D(x_i)$ in $\operatorname{Spec} R$, and the collection of these form an open cover of U. The projection $\pi|_{D(x_i)} : D(x_i) \to D_+(x_i)$ is induced by the inclusion $(R_{x_i})_0 \subseteq R_{x_i}$. The key point is that R_{x_i} decomposes as an $(R_{x_i})_0$-modules as

$$R_{x_i} = \bigoplus_{d\in\mathbb{Z}} (R_{x_i})_0 \cdot x_i^d.$$

The restriction of $\pi^*\mathscr{F}$ to $D(x_i)$ is the sheaf associated to the R_{x_i}-module

$$\begin{aligned} \Gamma(D(x_i), \pi^*\mathscr{F}) =& \Gamma(D_+(x_i), \mathscr{F}) \otimes_{(R_{x_i})_0} R_{x_i} \\ =& \bigoplus_{d\in\mathbb{Z}} \Gamma(D_+(x_i), \mathscr{F}) \cdot x_i^d. \end{aligned} \tag{2.8}$$

Using the fact that $D_+(x_i)$ is affine, we obtain for each d,

$$\begin{aligned} \Gamma(D_+(x_i), \mathscr{F}(d)) &= \Gamma(D_+(x_i), \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{O}_X(d)) \\ &= \Gamma(D_+(x_i), \mathscr{F}) \otimes_{(R_{x_i})_0} \Gamma(D_+(x_i), \mathscr{O}_X(d)) \\ &= \Gamma(D_+(x_i), \mathscr{F}) \otimes_{(R_{x_i})_0} (R_{x_i})_0 \cdot x_i^d \\ &= \Gamma(D_+(x_i), \mathscr{F}) \cdot x_i^d \end{aligned} \tag{2.9}$$

Comparing (2.8) and (2.9), we obtain an isomorphism

$$\Gamma(D_+(x_i), \pi_*\pi^*\mathscr{F}) = \bigoplus_{d\in\mathbb{Z}} \Gamma(D_+(x_i), \mathscr{F}(d)). \tag{2.10}$$

The isomorphisms (2.10) induce isomorphisms

$$\pi_*\pi^*\mathscr{F}|_{D_+(x_i)} \xrightarrow{\sim} \bigoplus_{d\in\mathbb{Z}} \mathscr{F}(d)|_{D_+(x_i)}.$$

The isomorphisms over $D(x_i)$ and $D(x_j)$ restrict to the same isomorphism

$$\Gamma(D_+(x_ix_j), \pi_*\pi^*\mathscr{F}) = \bigoplus_{d\in\mathbb{Z}} \Gamma(D_+(x_ix_j), \mathscr{F}(d)),$$

over $D(x_i) \cap D(x_j)$, so they glue to the desired isomorphism $\pi_*\pi^*\mathscr{F} = \bigoplus_{d\in\mathbb{Z}} \Gamma(D_+(x_ix_j), \mathscr{F}(d))$. □

As a byproduct of the above proof, we see that $M = \Gamma(U, \pi^*\mathscr{F})$ has the structure of a graded R-module, with degree d-component equal to $\Gamma(X, \mathscr{F}(d))$. Hence we may consider the sheaf $\widetilde{M}$ on X.

We claim that $\mathscr{F} \cong \widetilde{M}$. To show this, it suffices to show that the two sheaves agree over each $D_+(f)$ where $f \in R$ is homogeneous of positive degree. For such an f, we have

$$\begin{aligned}
\Gamma(D_+(f), \widetilde{M}) &= \Gamma(D(f), \widetilde{M})_0 \\
&= \Gamma(D(f), \iota_*\pi^*\mathscr{F})_0 \\
&= \Gamma(D(f), \pi^*\mathscr{F})_0 \\
&= \Gamma(D_+(f), \pi_*\pi^*\mathscr{F})_0 \\
&= \Gamma(D_+(f), \bigoplus_{d\in\mathbb{Z}} \mathscr{F}(d))_0 \\
&= \Gamma(D_+(f), \mathscr{F}).
\end{aligned}$$

These isomorphisms are compatible with restrictions to $D_+(fg)$ so they define an isomorphism $\widetilde{M} \cong \mathscr{F}$. □

Corollary 2.3.1

Let $X \subseteq \mathbb{P}^n_A$ be a projective scheme. If $\mathscr{F}$ is an $\mathscr{O}_X$-module of finite type, then there exists an integer $d > 0$ and a surjective map of $\mathscr{O}_X$-modules

$$\mathscr{O}_X(-d)^{\bigoplus N} \longrightarrow \mathscr{F}.$$

Proof By Theorem 2.3.1, there exists a finitely generated graded R-module M, such that $\mathscr{F} \cong \widetilde{M}$. Since M is finitely generated, there exists a finite set of elements $\{m_1, \ldots, x_N\}$ that generate M as an R-module. Because M is a graded module, we can always choose this finite generating set to consist of homogeneous elements. Let the degree of each generator be $\deg(m_i) = d_i$ for $i = 1, \ldots, N$. Define a free graded R-module L with a basis of N elements, $\{e_1, \ldots, e_N\}$, where we assign the degree of each basis element to match the degree of the corresponding generator of M:

$$\deg(e_i) = d_i.$$

This free graded module L can be expressed as a direct sum of twisting modules:

$$L = \bigoplus_{i=1}^{N} R(-d_i).$$

(Recall that $R(-d_i)$ is the free R-module of rank 1 generated by an element of degree d_i.)

Now, we can define a homomorphism of graded R-modules $\varphi : L \to M$ by specifying where the basis elements go:

$$\varphi(e_i) = m_i.$$

Since the set $\{m_1, \ldots, m_N\}$ generates M, this map φ is by construction surjective, i.e., $\bigoplus_{i=1}^{N} R(-d_i) \twoheadrightarrow M$. Let

$d = \max_i\{d_i\}$. For each generator m_i, we can create a new set of generators for M that all have degree d. Since R is generated by degree 1 elements, we can consider the set of elements: $\{x_j^{d-d_i} \cdot m_i\}_{i=1,\ldots,N; j=0,\ldots,n}$. This is a new finite set of homogeneous elements, all of which have degree d. This set also generates M. Call these new generators $\{m'_1, \ldots, m'_{N'}\}$. Now, we can construct a new free module $L' = \bigoplus_{j=1}^{N'} R(-d)$ and a new surjective map $\varphi' : L' \to M$. Recall that functor $\widetilde{(-)}$ is right exact, we have

$$\widetilde{R(-d)}^{\bigoplus N'} \longrightarrow\!\!\!\!\!\rightarrow \widetilde{M},$$

then we are done. □

To summarize, we have constructed two functors, we have constructed two functors,

$$\widetilde{(-)} : \mathbf{GrMod}_R \to \mathbf{QCoh}_X \quad \text{and} \quad \Gamma_* : \mathbf{QCoh}_X \to \mathbf{GrMod}_R\,.$$

Since $\widetilde{(\Gamma_*(\mathscr{F}))} \xrightarrow{\sim} \mathscr{E}$ is an isomorphism, it follows that $\widetilde{\square}$ is essentially surjective, that is, every quasicoherent sheaf on X is isomorphic to one of the form $\widetilde{M}$. However, unlike the affine case, the functors do not give an equivalence. The functor $\widetilde{\square}$ is not faithful, as the tilde of any module M with support in $V_+(R_+)$ is the zero sheaf. By Lemma 2.1.1 however, this is the only source of ambiguity.

Example 2.4 For $X = \mathbb{P}^n_A$, $n \geq 1$, Proposition 1.6.1 show that $\Gamma_*(\mathscr{O}_X) = A[x_0, \ldots, x_n]$.

When R is not a polynomial ring, it can easily happen that $\Gamma_*(\mathscr{O}_X)$ is different than R. A silly example is given by $X = \operatorname{Proj} k[t] = \operatorname{Spec} k$. Then $\Gamma(X, \mathscr{O}_X(1)) = \Gamma(D_+(t), \mathscr{O}_X(1)) = (k[t](1)_t)_0 = (k[t, t^{-1}](1))_0 = k[t, t^{-1}]_1 = kt$, so $\mathscr{O}_X(1) \cong \mathscr{O}_{\operatorname{Spec} k}$. Hence, we have $\mathscr{O}_X(m) \cong \mathscr{O}_X(1)^{\otimes n} \cong \mathscr{O}_{\operatorname{Spec} k}^{\otimes n} \cong \mathscr{O}_{\operatorname{Spec} k}$ for all m, so $\Gamma(X, \mathscr{O}_X(m)) = k$ for all m, even $m < 0$. $\Gamma(X, \mathscr{O}_X(m)) = k$ for all m, even $m < 0$. Hence $\Gamma_*(\mathscr{O}_X) = \bigoplus_d \Gamma(X, \mathscr{O}_X(d)) = k^{\oplus \mathbb{Z}} \cong k[t, t^{-1}] \ncong k[t]$.

Chapter 3 Divisors

Preliminaries

Function field of an integral scheme

Definition 3.0.1 (Function field)

Let X be an integral scheme. Then the local ring $\mathscr{O}_{X,\eta}$ of the generic point of X is a field, called the function field of X, and is denoted by $K(X)$.

For any affine open subset $U = \operatorname{Spec} R$ of X, we have $K(X) = \operatorname{Frac}(R)$.

Dimension

Definition 3.0.2 (Dimension and codimension)

(1) *The dimension of a scheme X is defined as $\dim X = \dim \operatorname{Sp}(X)$.*

(2) *If $Z \subseteq X$ is an irreducible closed subset, then the **codimension** of Z, denoted by $\operatorname{codim}_X(Z)$, is defined as the supremum of the length of chains of irreducible closed subsets*

$$Z = Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_l.$$

(3) *For any $Y \subseteq X$, define $\operatorname{codim}_X(Y) := \inf\limits_{\text{irr.cl. } Z \subseteq Y} \operatorname{codim}_X(Z)$, where the infimum is taken over all irreducible closed subsets Z of X with $Z \subseteq Y$.*

Proposition 3.0.1

Let X be an integral scheme of finite type over k. Then

(1) $\dim X = \dim \mathscr{O}_{X,x}$ *for any closed point $x \in X$,*

(2) $\dim X = \dim U$ *for any nonempty open subset $U \subseteq X$,*

(3) $\dim X = \operatorname{tr.deg}_k K(X)$,

(4) $\operatorname{codim}_X(Y) + \dim Y = \dim X$ *for any closed subset $Y \subseteq X$, and*

(5) $\operatorname{codim}_X(Y) = \inf\{\dim \mathscr{O}_{X,x} : x \in Y\}$ *for any closed subset $Y \subseteq X$.*

Discrete valuation rings

Definition 3.0.3

*Let K be a field. A **discrete valuation** is a surjective function $\nu : K \to \mathbb{Z} \cup \{\infty\}$ such that, for each $f, g \in K$*

(i) $\nu(fg) = \nu(f) + \nu(g)$.

(ii) $\nu(f + g) \geq \min\{\nu(f), \nu(g)\}$, *with equality in the latter when $\nu(f) \neq \nu(g)$.*

(iii) $\nu(f) = \infty$ *if and only if $f = 0$.*

Definition 3.0.4 (Discrete valuation rings)

*An integral domain A with fraction field K is called a **discrete valuation ring** or a **DVR** if there exists a* *valuation $\nu : K \to \mathbb{Z} \cup \{\infty\}$ such that*

$$A = \{x \in K : \nu(x) \geq 0\}.$$

Given a valuation ν, the set $\{x \in K : \nu(x) \geq 0\}$ is a subring of K. Therefore the discrete valuation ring determines and is determined by ν.

Proposition 3.0.2

Let A be a DVR with fraction field K. Then

(i) A is a local ring with maximal ideal

$$\mathfrak{m} = \{f \in K : \nu(f) \geq 1\}$$

and group of units

$$A^\times = \{f \in K : \nu(f) = 0\}.$$

(ii) If $t \in A$ is an element with $\nu(t) = 1$, then $\mathfrak{m} = (t)$.

(iii) Every nonzero ideal of A is of the form (t^n) for some $n \geq 1$ and $\nu(t) = 1$.

(iv) A is Noetherian of Krull dimension 1.

(v) A is normal.

Proposition 3.0.3

Let A be a Noetherian local domain of dimension 1 with maximal ideal $\mathfrak{m}$. Then the following are equivalent:

(i) A is DVR.

(ii) The maximal ideal $\mathfrak{m}$ is principal.

(iii) Every nonzero ideal is a power of $\mathfrak{m}$.

(iv) A is integrally closed.

(v) A is regular, i.e., $\dim_k \mathfrak{m}/\mathfrak{m}^2 = 1$.

3.1 Weil divisors

Note *We assume that X is a separated Noetherian integral normal scheme.*

Definition 3.1.1 (Prime divisor)

*A **prime divisor** is an integral subscheme $Z \subseteq X$ of codimension 1.*

Definition 3.1.2 (Weil divisor)

*A **Weil divisor** (or simply divisor), is a finite formal sum*

$$D = m_1 Z_1 + \cdots + m_r Z_r,$$

where $m_i \in \mathbb{Z}$ and the Z_i's are prime divisors. The set of divisors form a group, which will be denoted by $\mathrm{Div}(X)$.

*We say that a divisor D is **effective** if each $m_i \geq 0$, and we write $D \geq 0$. The **support** of D is defined to be the closed subset $\mathrm{Supp}(D) = Z_1 \cup \cdots \cup Z_r$.*

Example 3.1 If $X = \operatorname{Spec} A$ is affine, then every prime divisor is of the form $V(\mathfrak{p})$ where $\mathfrak{p}$ is a prime ideal of height 1. Hence $\operatorname{Div}(\operatorname{Spec} A) = \bigoplus_{\operatorname{ht}\mathfrak{p}=1} \mathbb{Z}\cdot V(\mathfrak{p})$ can be identified with the free abelian group on all height 1 prime ideals.

Example 3.2 On $X = \mathbb{P}^1_{\mathbb{C}}$, the prime divisors are simply the closed points. Here are some examples of Weil divisors:

$$\begin{aligned} D_1 &= 3\cdot[1:0] - 5\cdot[0:1],\\ D_2 &= [1:1] + 5\cdot[0:1],\\ D_1 + D_2 &= 3\cdot[1:0] + [1:1]. \end{aligned}$$

The reader may wonder why we include the assumption that X is normal. Certainly, the above definition can be made for any scheme, but the concept of a Weil divisor is not particularly useful without the normality assumption. The main reason is that there is a well-behaved notion of "order of vanishing", or "multiplicity" for rational functions.

More precisely, as the scheme X is normal, all the the local rings $\mathscr{O}_{X,x}$ are integrally closed in their fraction field, $K(X)$. This means that if $Z \subseteq X$ is a prime divisor, and $\zeta \in X$ is the generic point of Z, then the local ring $\mathscr{O}_{X,\zeta}$ is a 1-dimensional normal ring, and hence a DVR (Proposition 3.0.3). We write

$$\operatorname{ord}_Z : K(X) \longrightarrow \mathbb{Z}\cup\{\infty\}$$

for the corresponding valuation. More explicitly, the function ord_Z can be defined as follows: Given a nonzero $f \in \mathscr{O}_{X,\zeta}$, we can write it as $f = u\cdot t^m$, where t is the generator for maximal ideal of $\mathscr{O}_{X,\zeta}$ and u is a unit, and define the valuation of f at Z to be $\operatorname{ord}_Z(f) = m$. Finally, we extend this to elements of $K(X)$ by setting $\operatorname{ord}_Z(f/g) = \operatorname{ord}_Z(f) - \operatorname{ord}_Z(g)$ and $\operatorname{ord}_Z(0) = \infty$. The number $\operatorname{ord}_Z(f)$ is called the **order of vanishing of f at Z**. If $\operatorname{ord}_Z(f)$ is non-negative, we say that f **vanishes to that order along** Z. If $\operatorname{ord}_Z(f)$ is negative, then we say that f has a **pole** of that order along Z.

The following theorem is useful, sometimes called the Algebraic Hartogs Lemma:

Theorem 3.1.1 (Algebraic Hartogs Lemma)

Let R be a normal Noetherian domain. Then

$$R = \bigcap_{\operatorname{ht}\mathfrak{p}=1} R_{\mathfrak{p}},$$

where the intersection is taken over all $\mathfrak{p} \in \operatorname{Spec} R$ of height 1 and all rings are consider as subrings of $K = \operatorname{Frac}(R)$.

If $X = \operatorname{Spec} A$ is an affine scheme, then all prime divisors are of the form $V(\mathfrak{p})$ where $\mathfrak{p}$ is a prime ideal of height 1. For such a prime $\mathfrak{p}$, the local ring $A_{\mathfrak{p}}$ is a discrete valuation ring, consisting of exactly the fractions $f = a/b \in K(A)$ where $b \notin \mathfrak{p}$, or equivalently, $\operatorname{ord}_{V(\mathfrak{p})}(f) \geq 0$. By the Algebraic Hartogs Lemma 3.1.1, which says that $A = \bigcap_{\operatorname{ht}\mathfrak{p}=1} A_{\mathfrak{p}}$, we conclude that $\operatorname{ord}_{V(\mathfrak{p})}(f) \geq 0$ for every prime divisor $V(\mathfrak{p})$ if and only if $f \in A$. Moreover, $f \in K(A)$ has zero order of vanishing along every prime divisor if and only if f is invertible in $A^\times$.

In general, for an integral scheme X, a rational function $f \in K(X)$ is **regular** if and only if $f \in \mathscr{O}_{X,x}$ for every $x \in X$, which by the affine case above means that $\operatorname{ord}_Z(f) \geq 0$ for every prime divisor Z. Therefore, we have:

Proposition 3.1.1

For a rational function $f \in K(X)^\times$, we have

(i) $\operatorname{ord}_Z(f) \geq 0$ *for all prime divisors $Z \subseteq X$ if and only if $f \in \mathscr{O}_X(X)$.*

(ii) $\operatorname{ord}_Z(f) = 0$ *for all prime divisors* Z *if and only if* $f \in \mathscr{O}_X(X)^\times$.

This is illustrates one of the close links between rational functions and codimension 1 subschemes on normal schemes. If a rational function is regular outside a closed subset of codimension at least 2, then it is regular everywhere. (Sketch of proof: Let $Y \subseteq X$ closed subset with $\operatorname{codim}_X(Y) \geq 2$. For any prime divisor Z, we have $Z \nsubseteq Y$, if not, $\operatorname{codim}_X(Y) \leq \operatorname{codim}_X(Z) = 1$, a contradiction. Consider $U = X \setminus Y$, note that $Z \cap U$ is an irreducible closed subset of U, say $Z \cap U = \operatorname{cl}_U(\eta_Z)$ where η_Z is generic point, since f regular at $X \setminus Y$, $f \in \mathscr{O}_{X,\eta_Z}$. Since $\mathscr{O}_{X,\eta_Z}$ is DVR for all prime divisor Z, $\operatorname{ord}_Z(f) \geq 0$, by Proposition 3.1.1, f is regular everywhere.)

Definition 3.1.3 (Principal divisor)

For a rational function $f \in K(X)^\times$*, we define its corresponding divisor by*

$$\operatorname{div}(f) = \sum_{\text{prime divisor } Z} \operatorname{ord}_Z(f) Z, \tag{3.1}$$

where the sum runs over all prime divisors. Divisors of the form $\operatorname{div}(f)$ *are called* ***principal divisors****, and they form a subgroup of* $\operatorname{Div}(X)$.

To ensure that this is well-defined, we need to verify that for a nonzero rational function $f \in K(X)$, there are only finitely many prime divisors $Z \subseteq X$ such that $\operatorname{ord}_Z(f) \neq 0$:

Proposition 3.1.2

Let X *be a separated Noetherian integral normal scheme, and* $f \in K(X)^\times$*. Then there are only finitely many prime divisors* $Z \subseteq X$ *such that* $\operatorname{ord}_Z(f) \neq 0$.

Proof Let $U = \operatorname{Spec} A$ be any open affine subset such that $f|_U \in \mathscr{O}_X(U)$. For a prime divisor $Z \subseteq X$, there are two possible cases: (i) $Z \subseteq X \setminus U$, or (ii) $Z \cap U$ is a prime divisor of U. As we assume X is Noetherian and integral, there can be only finitely many codimension 1 components of the closed subset $X \setminus U$, so there are only finitely many Z's in case (i). For the ones in case (ii), note that $\operatorname{ord}_Z(f) \geq 0$ automatically, because f is regular in U, and $\operatorname{ord}_Z(f) > 0$ is and only if f vanishes along Z. Again because X is Noetherian, the zero set $V(f)$ has only finitely many components, so we conclude. □

Note that if f is a regular function, i.e., $f \in \mathscr{O}_X(X)$, then $\operatorname{div}(f)$ is an effective divisor. It is supported along the zero set of f, with multiplicity $\operatorname{ord}_Z(f)$ along each component Z.

Example 3.3 On $\operatorname{Spec} \mathbb{Z}$, a Weil divisor is an expression of the form

$$D = n_1 V(p_1) + \cdots + n_r V(p_r)$$

where the p_i are prime numbers. In the function field $\mathbb{Q}$, consider the rational function $f = p_1^{n_1} \cdots p_r^{n_r}$. Note that $f = p_i^{n_i} \cdot p_1^{n_1} \cdots \widehat{p_i^{n_i}} \cdots p_r^{n_r}$ in $\mathbb{Z}_{(p_i)}$, we have $\operatorname{ord}_{V(p_i)}(f) = n_i$ for all i. So $\operatorname{div}(f) = D$. Hence every divisor is principal on $\operatorname{Spec} \mathbb{Z}$.

Example 3.4 Let k be an algebraically closed field, and let $\mathbb{A}^1_k = \operatorname{Spec} k[t]$ be the affine line with fraction field $k(t)$. The prime divisor of $\mathbb{A}^1_k$ correspond to k-points $a \in \mathbb{A}^1_k$, each associated to maximal ideal $(t-a)$ in $k[t]$.

Consider the rational function

$$f = t^2(t-1)^{-1} \in k(t).$$

Note that for $a \in k$, the elements t and $(t-1)$ are invertible in the local ring $\mathscr{O}_{\mathbb{A}^1_k,a} = k[t]_{(t-a)}$ except when $a = 0$ or $a = 1$.

When $a = 0$, then t is a local parameter in $\mathscr{O}_{\mathbb{A}^1_k,a} = k[t]_{(t)}$, and we may write $f = t^2(\text{unit})$. Hence the order of vanishing of f is equal to 2 at the point $a = 0$. Similarly, $\text{ord}_a(f) = -1$ at $a = 1$. Hence

$$\text{div}(f) = 2V(t) - V(t-1).$$

More generally, if $f(t) = (t-a_1)^{n_1} \cdots (t-a_r)^{n_r}$, then

$$\text{div}(f) = n_1 V(t-a_1) + \cdots + n_r V(t-a_r). \tag{3.2}$$

In particular, every divisor D is the divisor of some nonzero rational function.

Example 3.5 The same rational function $f = t^2(t-1)^{-1}$ may be considered as a rational on $\mathbb{P}^1_k$, when $\mathbb{A}^1_k$ is embedded as the open set U_0 with coordinate $t = x_{1/0}$. We have already computed the values $\text{ord}_P(f)$ for $P \in U_0$. For the remaining point, $P = [0:1]$, we use the coordinate $s = x_{0/1} = t^{-1}$ and rewrite f as

$$(s^{-1})^2(s^{-1}-1)^{-1} = s^{-1}(1-s)^{-1}.$$

From this, we see that $\text{ord}_{[0:1]}(f) = -1$ and $\text{ord}_{[1:1]}(f) = -1$ (which was already computed in U_0). Therefore, the divisor of f on $\mathbb{P}^1_k$ is given by

$$\text{div}(f) = 2[1:0] - [1:1] - [0:1].$$

Example 3.6 Let X be the curve $V(x^3 - y^3 + y) \subseteq \mathbb{A}^2_k$. Then x, y and y/x^2 define rational functions on X. Here x and y are regular, so they have nonnegative orders every where. Let us find the points $P \in X$ where $\text{ord}_P(x) > 0$.

The function x vanishes exactly at the points in $V(x, x^3 - y^3 + y) \subseteq \mathbb{A}^2$, i.e., the three points $(0,0), (0,1), (0,-1)$. The local ring at the origin $(0,0)$ is isomorphic to

$$\mathscr{O}_{X,(0,0)} = (k[x,y]/(x^3 - y^3 + y))_{(x,y)}.$$

In this ring we have $x^3 = y(y^2-1)$, and so $y = x^3(unit)$. Hence x is a local parameter. In particular, $\text{ord}_{(0,0)}(x) = 1$ and $\text{ord}_{(0,0)}(y) = 3$. Similar computations show that

$$\text{div}(x) = (0,0) + (0,-1) + (0,1).$$

As for y, this can only have nonzero orders of vanishing at the points in $V(x^3 - y^3, y) = V(x, y)$, i.e., at the origin $(0,0)$. We have already computed that $\text{ord}_{(0,0)}(y) = 3$, so $\div(y) = 3(0,0)$. Putting everything together, we obtain

$$\text{div}(y/x^2) = 3(0,0) - 2((0,0) + (0,-1) + (0,1)) = (0,0) - 2(0,-1) - 2(0,1).$$

3.2 The class group

Note *We assume that X is a separated Noetherian integral normal scheme.*

One of the fundamental invariants of a scheme (or of a ring) is the class group, along with its close relative, the Picard group. The class group can be interpreted as the group that captures the obstructions to a divisor being the divisor of a rational function.

Definition 3.2.1 (Class group)

*The **class group** of X is defined as the group of Weil divisors modulo the principal divisor, i.e.,*

$$\text{Cl}(X) = \text{Div}(X)/(\text{div}(f) : f \in K(X)^\times).$$

*Two Weil divisors D and D' are said to be **linearly equivalent** (written $D \sim D'$) if they have the same image in $\text{Cl}(X)$, or equivalently, that $D - D'$ is principal.*

For an affine scheme $X = \operatorname{Spec} A$, the divisor of a rational function $f \in K(X)$ is equal to

$$\operatorname{div}(f) = \sum_{\operatorname{ht}\mathfrak{p}=1} \operatorname{ord}_{\mathfrak{p}}(f)V(\mathfrak{p}).$$

Hence $\operatorname{div}(f) = 0$ if and only if $\operatorname{ord}_{\mathfrak{p}}(f) = 0$ for all height 1 primes, which by Proposition 3.1.1 happens if and only if $f \in A^{\times}$. Hence the kernel of the map $\operatorname{div} : K(X)^{\times} \to \operatorname{Div}(\operatorname{Spec} A)$ equals $A^{\times}$, and the cokernel is by definition the class group $\operatorname{Cl}(\operatorname{Spec} A)$. Hence we have the exact sequence

$$0 \longrightarrow A^{\times} \longrightarrow K(X)^{\times} \xrightarrow{\operatorname{div}} \operatorname{Div}(\operatorname{Spec} A) \longrightarrow \operatorname{Cl}(\operatorname{Spec} A) \longrightarrow 0.$$

Example 3.7 The class group of $\operatorname{Spec}\mathbb{Z}$ is trivial, as $\mathbb{Z}$ is a principal ideal domain, so every prime ideal is principal, not just the ones of height 1. (See Example 3.3 .)

Example 3.8 The class group of $\operatorname{Spec}\mathbb{Z}[i]$ is also trivial, since every prime ideal of $\mathbb{Z}[i]$ is principal.

Example 3.9 Let A be a DVR and let $X = specA$. Recall Proposition 3.0.3, in A, the only nonzero prime ideal is the maximal ideal $\mathfrak{m}$. Therefore, if $x \in X$ denotes the closed point, we have $\operatorname{Div}(X) = \mathbb{Z}\cdot x$. Any Weil divisor on X is principal: if t is a generator for $\mathfrak{m}$, then $\operatorname{div}(t^n) = n \cdot x$ for each $n \in \mathbb{Z}$. Hence $\operatorname{Cl}(X) = 0$.

Both the terms 'divisor' and 'class group' originate from algebraic number theory and their origins can be traced back to Kummer's work on Fermat's Last Theorem. In this setting, the class groups are designed to measure how far the base ring is from being a unique factorization domain.

Proposition 3.2.1

Let A be a normal Noetherian integral domain. Then the following are equivalent:

(i) $\operatorname{Cl}(\operatorname{Spec} A) = 0$.

(ii) Every height 1 prime ideal in A is a principal ideal.

(iii) A is a UFD.

Proof The equivalence of (ii) and (iii) is a fact from commutative algebra. It remains to show the equivalence (i) $\Leftrightarrow$ (ii).

(ii) $\Rightarrow$ (i): Write $X = \operatorname{Spec} A$. If Z is a prime divisor in X, then $Z = V(\mathfrak{p})$ for some prime ideal $\mathscr{P} \subseteq A$, and as Z has codimension 1, $\mathscr{P}$ is of height 1. By assumption (ii), $\mathfrak{p} = (f)$ for an element $f \in A$, that is, $1 \cdot Z = \operatorname{div}(f)$. This shows that $\operatorname{Cl}(\operatorname{Spec} A) = 0$.

(i) $\Rightarrow$ (ii): Assume that $\operatorname{Cl}(\operatorname{Spec} A) = 0$. Let $\mathfrak{p}$ be a prime of height 1, and let $Z = V(\mathfrak{p}) \subseteq X$. By assumption, there is an $f \in K(X)^{\times}$ such that $\operatorname{div}(f) = Z$. We want to show that in fact $f \in A$ and that $\mathfrak{p} = (f)$. For the first statement: as $\operatorname{div}(f) = Z$, the valuation of f are given by $\operatorname{ord}_{\mathfrak{q}}(f) = 0$ for $\mathfrak{q} \neq \mathfrak{p}$ and $\operatorname{ord}_{\mathfrak{p}}(f) = 1$, by Proposition 3.1.1, $f \in \mathscr{O}_X(X) = A$.

Secondly, to prove that f generates $\mathfrak{p}$, consider any nonzero element $g \in \mathfrak{p}$. Then $\operatorname{ord}_{\mathfrak{p}}(g) \geq 1$, and by Proposition 3.1.1, $\operatorname{ord}_{\mathfrak{q}}(g) \geq 0$ for all $\mathfrak{q} \neq \mathfrak{p}$. It follows that $\operatorname{ord}_{\mathfrak{q}}(g/f) = \operatorname{ord}_{\mathfrak{q}}(g) - \operatorname{ord}_{\mathfrak{q}}(g) \geq 0$ for every prime ideal $\mathfrak{q} \in \operatorname{Spec} A$. Hence $g/f \in A_{\mathfrak{q}}$ for every prime $\mathfrak{q}$ of height 1, and hence $g/f \in A$, by Proposition 3.1.1. It follows that $g \in fA$, i.e., $g \in (f)$. Hence $\mathfrak{p} = (f)$ is principal. □

Example 3.10 The class group of affine n-space $\mathbb{A}^n_k = \operatorname{Spec} k[x_1, \ldots, x_n]$ is trivial, as the polynomial ring is a UFD.

Theorem 3.2.1

Let k be a field. The class group of projective n-space is

$$\operatorname{Cl}(\mathbb{P}^n_k) = \mathbb{Z},$$

generated by the class of a hyperplane.

Proof Let $x_0, \ldots, x_n$ be homogeneous coordinates and let $H = V_+(x_0)$ be the hyperplane defined by x_0. First we show that H generates $\mathrm{Cl}(\mathbb{P}_k^n)$. Let $Y \subseteq \mathbb{P}_k^n$ be any prime divisor. Then $Y = V_+(\mathfrak{a})$ for some homogeneous prime ideal of height 1 in R. As the polynomial ring $k[x_0, \ldots, x_n]$ is a UFD, we may write $\mathfrak{a} = (F)$ for some irreducible homogeneous polynomial F. Letting $d = \deg F$, the fraction $f = F/x_0^d$ defines a regular function on U_0, hence a rational function on $\mathbb{P}_k^n$. Note that f can only have nonzero orders of vanishing along Y and H. Since F is irreducible, we have $\operatorname{div}(F/x_0^d) = Y - dH$. Therefore $[Y] = dH$ in $\mathrm{Cl}(\mathbb{P}_k^n)$.

To conclude, we show that no nonzero multiple $[H]$ is zero in $\mathrm{Cl}(\mathbb{P}_k^n)$. If $m[H] = 0$ for some $m > 0$, then there would exist an $f \in K(\mathbb{P}_k^n)$ such that $mH = \operatorname{div}(f)$. Note that $\operatorname{div}(f)$ is effective, meaning $\operatorname{ord}_Z(f) \geq 0$ for every prime divisor Z. By Proposition 3.1.1, $f \in \mathscr{O}_{\mathbb{P}_k^n}(\mathbb{P}_k^n)$, hence f is a constant, which implies that $m = 0$.
□

The isomorphism $\mathrm{Cl}(X) \xrightarrow{\sim} \mathbb{Z}$ is better defined by the **degree map**. For a divisor $D = \sum n_i V_+(G_i)$, where $G_i \in k[x_0, \ldots, x_n]$ are irreducible homogeneous polynomials, we define

$$\deg D = \sum_i n_i \deg G_i.$$

As rational functions on $\mathbb{P}_k^n$ have the form F/G where $F, G \in k[x_0, \ldots, x_n]$ of the same degree, this descends to a map $\mathrm{Cl}(\mathbb{P}_k^n) \to \mathbb{Z}$.

3.3 The class group of an open set

Note *We assume that X is a separated Noetherian integral normal scheme.*

If $U \subseteq X$ is an open set and $Z \subseteq X$ is a prime divisor, the intersection $Z \cap U$ is either empty (if $Z \subseteq X - U$) or prime divisor on U. This allows us to define the restriction of a divisor $D = \sum n_Z Z$ to U by the formula

$$D|_U = \sum_{Z \cap U \neq \varnothing} n_Z \cdot Z \cap U. \tag{3.3}$$

This defines a map of groups $\operatorname{Div}(X) \to \operatorname{Div}(U)$. Note that any divisor with support in $X - U$ is sent to 0 via this map.

If f is a rational function on X, the restriction $f|_U$ is a rational function on U, and it holds that $\operatorname{ord}_{Z \cap U}(f|_Z) = \operatorname{ord}_Z(f)$ whenever $Z \cap U \neq \varnothing$ (the two valuation rings are the same). Consequently the divisor $\operatorname{div}(f)$ restricts to the divisor $\operatorname{div}(f|_U)$, and the restriction map sends principal divisors to principal divisors.

In this setting, it is natural to ask how the class group of X and U are related. The answer is given by the theorem below.

Theorem 3.3.1

Let X be a normal integral Noetherian scheme. Let $W \subseteq X$ be a closed subscheme and let $U = X \setminus W$. If $Z_1, \ldots, Z_r$ are the prime divisors corresponding to the codimension 1 components of W, there is an exact sequence

$$\bigoplus_{i=1}^r \mathbb{Z}Z_i \longrightarrow \mathrm{Cl}(X) \longrightarrow \mathrm{Cl}(U) \longrightarrow 0, \tag{3.4}$$

where the map $\mathrm{Cl}(X) \to \mathrm{Cl}(U)$ is defined by $[Z] \mapsto [Z \cap U]$.

Proof If Z is a prime divisor on U, the closure in X is a prime divisor in X, so the right-most map in (3.4) is surjective, and we just need to check exactness in the middle.

Suppose Z is a prime divisor which is principal on U. Then $Z|_U = \operatorname{div}(f)$ for some $f \in K(U) = K(X)$. Now $D = \operatorname{div}(f)$ is a divisor on X such that $D|_U = \operatorname{div}(f)|_U = Z|_U$. Hence $D - Z$ is a Weil divisor supported in $X \setminus W$, and hence it must be a linear combination of the Z_i's. Therefore $D - Z$ is in the image of the left-most map, and we done. □

Corollary 3.3.1

If Z is a closed subset of codimension at least 2, then

$$\mathrm{Cl}(X \setminus Z) = \mathrm{Cl}(X).$$

Proof Recall that $\operatorname{codim}_X(Z) = \inf_{\text{irr.cl. } Y \subseteq Z} \operatorname{codim}_X(Z)$, since $\operatorname{codim}_X(Z) \geq 2$, we know that Z does not have codimension 1 component, so in (3.4) $\bigoplus_{i=1}^{r} \mathbb{Z}Z_i = 0$, it follows that $\mathrm{Cl}(X) \cong \mathrm{Cl}(X - Z)$. □

Example 3.11 If P denotes the origin in $\mathbb{A}_k^n$, by Proposition 3.0.1 (4), we have $\operatorname{codim}(P) = \dim \mathbb{A}_k^n - \dim P = n - 0 = n$. Hence if $n \geq 2$, by Corollary 3.3.1, $\mathrm{Cl}(\mathbb{A}_k^n \setminus \{P\}) = \mathrm{Cl}(\mathbb{A}_k^n)$. Since polynomial ring is UFD, by Proposition 3.2.1, $\mathrm{Cl}(\mathbb{A}_k^n) = 0$, and therefore $\mathrm{Cl}(\mathbb{A}_k^n \setminus \{P\}) = 0$.

If $n = 1$, we have $\mathrm{Cl}(\mathbb{A}_k^1 \setminus \{P\}) = \mathrm{Cl}(\operatorname{Spec} k[x]_x) = 0$ because $k[x, x^{-1}]$ is a UFD.

Example 3.12 Consider the projective line $\mathbb{P}_k^1$ over a field k, and let P be a k-point. Note that $\operatorname{codim} P = \dim \mathbb{P}_k^1 - \dim P = 1 - 0 = 1$, by Theorem 3.3.1, we have the exact sequence

$$\mathbb{Z} \cdot [P] \longrightarrow \mathrm{Cl}(\mathbb{P}_k^1) \longrightarrow \mathrm{Cl}(\mathbb{A}_k^1) \longrightarrow 0.$$

We saw that $\mathrm{Cl}(\mathbb{A}_k^1) = 0$, so the map $\mathbb{Z} \to \mathrm{Cl}(\mathbb{P}_k^1)$ is surjective. It is also injective: if $n[P] = [nP] = 0$ in $\mathrm{Cl}(\mathbb{P}_k^1)$ for some n, then $nP = \operatorname{div}(f)$ for some $f \in K(\mathbb{P}^1)$. We have $nP|_{\mathbb{A}_k^1} = 0$, so we must have $\operatorname{div}(f)|_{\mathbb{A}_k^1} = 0$. Therefore, f must have neither zeros nor poles on $\mathbb{A}_k^1$, implies that f is a constant, and hence $n = 0$.

The same proof works in higher dimensions, and gives a new proof of the formula $\mathrm{Cl}(\mathbb{P}_k^n) = \mathbb{Z}$.

Example 3.13 More generally, consider a fibered product $X = \mathbb{P}_k^{n_1} \times_k \cdots \times_k \mathbb{P}_k^{n_r}$. We claim that $\mathrm{Cl}(X) \cong \mathbb{Z}^r$. For each $i = 1, \dots, r$, let $h_i \subseteq \mathbb{P}_k^{n_i}$ be a hyperplane and consider the pullback $H_i = \mathrm{pr}_i^{-1}(h_i)$. Then we have

$$X - \bigcup_{i=1}^{r} H_i = \bigcap_{i=1}^{r} (X \setminus H_i) \cong \prod_{i=1}^{r} \mathbb{A}_k^{n_i} \cong \mathbb{A}_k^N.$$

where $N = \sum_i n_i$. Since this is an affine space, $\mathrm{Cl}(\mathbb{A}_k^N) = 0$, and hence the $H_1, \dots, H_r$ generate the class group of X. We claim that these classes are independent, i.e.,

$$\mathrm{Cl}(X) = \mathbb{Z}[H_1] \oplus \cdots \oplus \mathbb{Z}[H_r].$$

Indeed, if $\sum_{i=1}^{r} n_i H_i = \operatorname{div}(f)$ for some $f \in K(X)$, then $\operatorname{Supp}(\operatorname{div}(f)) \subseteq \bigcup_{i=1}^{r} H_i$, hence $\operatorname{ord}_Z(f) = 0$ for every Z which intersects $X \setminus \bigcup_{i=1}^{r} H_i$. Therefore, by Proposition 3.1.1, f is a unit in $k[x_1, \dots, x_N]$ and hence a constant. Hence $\operatorname{div}(f) = 0$, and hence $a_1 = \cdots = a_r = 0$.

3.4 Cartier divisors

3.4.1 Basic concepts

If $f \in \mathscr{O}_X(X)$ is a nonzero regular function, then $\operatorname{div}(f)$ is an effective divisor with support along the zero set of f, and "multiplicity" $\operatorname{ord}_Z(f)$ along each component. Generally, we except that a codimension 1 closed subscheme can be defined by a single equation, at least locally. As we will see this intuition is often correct, at least when X has mild singularities.

Definition 3.4.1 (Sheaf of total quotient rings of structure sheaf)

*Let X be a scheme. For each open affine subset $U = \operatorname{Spec} A$, let $S(U) = \{a \in \Gamma(U, \mathscr{O}_X) : a \text{ is not zerodivisor}\}$, and let $K(U) := S(U)^{-1}\Gamma(U, \mathscr{O}_X)$ be the localization of $\Gamma(U, \mathscr{O}_X)$ by the multiplicative set $S(U)$. We call $K(U)$ the **total quotient ring of** $\Gamma(U, \mathscr{O}_X)$. $K(U)$ form a presheaf, whose associated sheaf of rings $\mathscr{K}_X$ we call the **sheaf of total quotient rings of** $\mathscr{O}_X$. On an arbitrary scheme, the sheaf $\mathscr{K}_X$ replaces the concept of function field of an integral scheme. We denote by $\mathscr{K}_X^*$ **the sheaf of multiplicative groups of invertible elements in the sheaf of rings** $\mathscr{K}_X$. Similarly, $\mathscr{O}_X^*$ is the sheaf of invertible elements in $\mathscr{O}_X$.*

Definition 3.4.2 (Cartier divisor)

*A **Cartier divisor** on a scheme X is a global section of the sheaf $\mathscr{K}_X^*/\mathscr{O}_X^*$, denote $\operatorname{Div}(X) := \Gamma(X, \mathscr{K}_X^*/\mathscr{O}_X^*)$. Let $f \in \Gamma(X, \mathscr{K}_X^*)$, its image in $\Gamma(X, \mathscr{K}_X^*/\mathscr{O}_X^*)$ is called a **principal Cartier divisor** and denoted by $\operatorname{div}(f)$. The group law on $\Gamma(X, \mathscr{K}_X^*/\mathscr{O}_X^*)$ is noted additively. We say that two Cartier divisors D_1 and D_2 are **linearly equivalent** if $D_1 - D_2$ is principal. We then write $D_1 \sim D_2$. A Cartier divisor D is called **effective** if it is in the image of the canonical map $\Gamma(X, \mathscr{O}_X \cap \mathscr{K}_X^*) \to \Gamma(X, \mathscr{K}_X^*/\mathscr{O}_X^*)$. We then write $D \geq 0$. The set of effective Cartier divisors is denoted by $\operatorname{Div}_+(X)$. Let U be an open subset of X, and let $D|_U$ denote the **restriction of** X (as a section of the sheaf $\mathscr{K}_X^*/\mathscr{O}_X^*$) to U.*

By definition, we can represent a Cartier divisor D by a system $\{(U_i, f_i)_i\}$ (we call it **Cartier data**), where the U_i are open subsets of X forming a covering of X, $f_i \in \Gamma(U_i, \mathscr{K}_X^*)$, and $f_i|_{U_i \cap U_j} \in f_j|_{U_i \cap U_j}\Gamma(U_i \cap U_j, \mathscr{O}_X^*)$ for every i, j. Two system $\{(U_i, f_i)_i\}$ and $\{(V_j, g_j)_j\}$ represent the same Cartier divisor if on $U_i \cap V_j$, f_i and g_j differ by a multiplicative factor in $\mathscr{O}_X^*(U_i \cap V_j)$. Let D_1, D_2 be two Cartier divisors, represented by $\{(U_i, f_i)_i\}$ and $\{(V_j, g_j)_j\}$, respectively. Then $D_1 + D_2$ is represented by $\{(U_i \cap V_j, f_i g_j)_{i,j}\}$. We have $D \geq 0$ if and only if can be represented by $\{(U_i, f_i)_i\}$ with $f_i \in \mathscr{O}_X(U_i)$. It is principal if it can be represented by a system $\{(X, f)\}$.

Example 3.14 Consider the projective n-space $\mathbb{P}_k^n$ over a field k. Write $\mathbb{P}_n^k = \operatorname{Proj} R$ where $R = k[x_0, \ldots, x_n]$. Any homogeneous polynomial of degree d, $F(x_0, \ldots, x_n) \in R_d$ defines a closed subscheme of $\mathbb{P}_k^n$ of codimension 1. The corresponding Weil divisor D is Cartier. Concretely, we can write down the Cartier data with respect to the standard covering $U_i = D_+(x_i)$ of $\mathbb{P}_k^n$. Note that $F\left(\frac{x}{x_i}\right) = F\left(\frac{x_0}{x_i}, \ldots, \frac{x_{i-1}}{x_i}, 1, \frac{x_{i+1}}{x_i}, \ldots, \frac{x_n}{x_i}\right)$ defines a nonzero regular function on U_i, and the collection

$$\left(U_i, F\left(\frac{x}{x_i}\right)\right)$$

forms a Cartier divisor D on X. Indeed, on the overlap $U_i \cap U_j$, we have the relation

$$F\left(\frac{x}{x_i}\right) = \left(\frac{x_j}{x_i}\right)^d F\left(\frac{x}{x_j}\right) \tag{3.5}$$

and $x_{j/i}$ is a regular and invertible function on $U_i \cap U_j$. Two homogeneous polynomials F, G of the same degree d give linearly equivalent divisors, because the quotient $\frac{F(x)}{G(x)}$ is a global rational function on $\mathbb{P}_k^n$.

3.4.2 Under good circumstances, Weil divisors = Cartier divisors

It is not true in general that every divisor is Cartier. However, if X is **factorial**, meaning that all the local rings $\mathscr{O}_{X,x}$ are UFD's, then the two notions are the same:

Proposition 3.4.1

Let X be an integral, separated noetherian scheme, all of whose local rings are UFDs (in which case we say X is locally factorial). Then the group $\mathrm{Div}(X)$ of Weil divisors on X is isomorphic to the group of Cartiers divisors $\Gamma(X, \mathscr{K}_X^*/\mathscr{O}_X^*)$, i.e., $\mathrm{Div}(X) \cong \Gamma(X, \mathscr{K}_X^*/\mathscr{O}_X^*)$, and furthermore, the principal Weil divisors correspond to the principal Cartier divisors under this isomorphism.

Proof Since all local rings of X are UFDs, and note that UFDs are integrally closed, X is normal. Since X is integral, the sheaf $\mathscr{K}$ is just the constant sheaf corresponding to the function field $K := K(X)$ of X. Now let a Cartier divisor be given by $\{(U_i, f_i)\}$ where $\{U_i\}$ is an open cover of X, and $f_i \in \Gamma(U_i, \mathscr{K}*) = K^*$. We define the associated Weil divisor as follows. For each prime divisor Z, take the coefficient of Z to be $\mathrm{ord}_Z(f_i)$, where i is any index for which $Z \cap U_i \neq \varnothing$. If j is another such index, then f_i/f_j is invertible on $U_i \cap U_j$ (since $f_i|_{U_i \cap U_j} \in f_j|_{U_i \cap U_j} \Gamma(U_i \cap U_j, \mathscr{O}_X^*)$), so $\mathrm{ord}_Z(f_i/f_j) = 0$ and $\mathrm{ord}_Z(f_i) = \mathrm{ord}_Z(f_j)$. Thus we obtain a well-defined Weil divisor $D = \sum \mathrm{ord}_Z(f_i)Y$ on X. (The sum is finite, since X is Noetherian.)

Conversely, if D is a Weil divisor on X, let $x \in X$ be any point. Then D induces a Weil divisor D_x on the local scheme $\mathrm{Spec}\, \mathscr{O}_{X,x}$. Since $\mathscr{O}_{X,x}$ is a UFD, by Proposition 3.2.1, D_x is a principal divisor, so let $D_x = \mathrm{div}(f_x)$ for some $f_x \in K$. Now the principal divisor $\mathrm{div}(f_x)$ on X has the same restriction to $\mathrm{Spec}\, \mathscr{O}_x$ as D, hence they differ only at prime divisors which do not pass through x. There are only finitely many of these which have a non-zero coefficient in D or $\mathrm{div}(f_x)$, so there is an open neighborhood U_x of such that D and $\mathrm{div}(f_x)$ have the same restriction to U_x. Covering X with such open sets U_x, the functions f_x give a Cartier divisor on X. Note that if f, f' give the same Weil divisor on an open set U, then $f/f' \in \Gamma(U, \mathscr{O}_X^*)$, since X is normal.

These two constructions are inverse to each other, so we see that the groups of Weil divisors and Cartier divisors are isomorphic. Furthermore it is clear that the principal divisors correspond to each other. □

Corollary 3.4.1

Let X be a Noetherian, regular scheme. Then every Weil divisor is Cartier.

Proof If X is regular, then each local ring $\mathscr{O}_{X,x}$ is a regular local ring, hence a UFD by the Auslander-Buchsbaum Theorem. □

Note *We denote by $\mathrm{CaCl}(X)$ the group of isomorphism classes of Cartier divisors modulo the linear equivalence relation.*

3.4.3 Sheaf associated to Cartier divisor

Definition 3.4.3 (Sheaf associated to Cartier divisor)

Let D be a Cartier divisor on a scheme X, represented by $\{(U_i, f_i)\}$ as above. We define a subsheaf $\mathscr{O}_X(D)$ of the sheaf of total quotient rings $\mathscr{K}$ by taking $\mathscr{O}_X(D)$ to be the sub-$\mathscr{O}_X$-module of $\mathscr{K}$ generated by f_i^{-1} on U_i, i.e.,

$$\mathscr{O}_X(D)|_{U_i} := \frac{1}{f_i} \cdot \mathscr{O}_X|_{U_i}.$$

This is well-defined, since f_i/f_j is invertible on $U_i \cap U_j$, so f_i^{-1} and f_j^{-1} generated the same $\mathscr{O}_X$-module. We call $\mathscr{O}_X(D)$ the sheaf associated to D.

Remark By construction, we see that $\mathscr{O}_X(D)$ is an invertible sheaf on X. And $D \geq 0$ if and only if $\mathscr{O}_X(-D) \subseteq \mathscr{O}_X$. If U is an open subset of X, then $\mathscr{O}_X(D)|_U = \mathscr{O}_U(D|_U)$.

Remark Let X be a Noetherian integral normal scheme by Proposition 3.4.1, the Cartier divisors on X are same as Weil divisors on X, let D be a divisor on X. Say $D = \{(U_i, f_i)\}$, by the proof of Proposition 3.4.1, $D = \sum\limits_{Z \cap U_i \neq \varnothing, \forall i} n_Z Z$ where $n_Z = \operatorname{ord}_Z(f_i)$. Then

$$\begin{aligned}
\mathscr{O}_X(D)(U_i) =& \frac{1}{f_i}\mathscr{O}_X(U_i) \\
=& \{s \in \mathscr{K}_X(U_i) : s = g/f_i, g \in \mathscr{O}_X(U_i)\} \\
=& \{s \in \mathscr{K}_X(U_i) : sf_i \in \mathscr{O}_X(U_i)\} \\
=& \{s \in \mathscr{K}_X(U_i) : \operatorname{ord}_Z(s) \geq -n_Z, \forall \text{ prime divisor } Z \text{ with } Z \cap U_i \neq \varnothing\}.
\end{aligned}$$

for all U_i. Hence for any Weil divisor $D = \sum n_Z Z$, we have

$$\mathscr{O}_X(D)(U) = \{s \in K(X) : \operatorname{ord}_Z(s) \geq -n_Z, \forall \text{ prime divisor } Z \text{ with } Z \cap U \neq \varnothing\}.$$

Equivalent, we can formulate it by saying that $\operatorname{div}(f) + D$ is an effective Weil divisor; in other words, $\operatorname{div}(f) + D \geq 0$. Informally, this means that the "pole part" of $\operatorname{div}(f)$ is canceled out by adding D.

Proposition 3.4.2

Let X be a scheme. Then:

(a) for any Cartier divisor D, $\mathscr{O}_X(D)$ is an invertible sheaf on X. The map $D \longmapsto \mathscr{O}_X(D)$ gives a one-to-one correspondence between Cartier divisors on X and invertible subsheaves of $\mathscr{K}_X$;

(b) $\mathscr{O}_X(D_1 - D_2) \cong \mathscr{O}_X(D_1) \otimes \mathscr{O}_X(D_2)^{-1}$;

(c) $D_1 \sim D_2$ if and only if $\mathscr{K}(D_1) \cong \mathscr{K}(D_2)$.

Proof (a). Since for each $f_i \in \Gamma(U_i, \mathscr{K}^*)$, the map $\mathscr{O}_{U_i} \to \mathscr{O}_X(D)|_{U_i}$ defined by $1 \mapsto f_i^{-1}$ is an isomorphism. Thus $\mathscr{O}_X(D)$ is an invertible sheaf. The Cartier divisor D can be recovered from $\mathscr{O}_X(D)$ together with its embedding in $\mathscr{K}$, by taking f_i on U_i to be the inverse of a local generator of $\mathscr{O}_X(D)$. For ant invertible subsheaf of $\mathscr{K}$, this construction gives a Cartier divisor, so we have an one-to-one correspondence as claimed.

(b). If D_1 is locally defined by f_i and D_2 is locally defined by g_i, then $\mathscr{O}_X(D_1 - D_2)$ is locally generated by $f_i^{-1}g_i$, so $\mathscr{O}_X(D_1 - D_2) = \mathscr{O}_X(D_1) \cdot \mathscr{O}_X(D_2)^{-1}$ as subsheaves of $\mathscr{K}$. This product is clearly isomorphic to the tensor product $\mathscr{O}_X(D_1) \otimes_{\mathscr{O}_X} \mathscr{O}_X(D_2)^{-1}$.

(c). Using (b), it will be sufficient to show that $D = D_1 - D_2$ is principal if and only if $\mathscr{O}_X(D) \cong \mathscr{O}_X$. If D is principal, defined by $f \in \Gamma(X, \mathscr{K}^*)$, then $\mathscr{O}_X(D)$ is globally generated by f^{-1}, so sending $1 \mapsto f^{-1}$ gives an isomorphism $\mathscr{O}_X \cong \mathscr{O}_X(D)$. Conversely, given such an isomorphism, the image of 1 in $\Gamma(X, \mathscr{O}_X)$

gives an element of $\Gamma(X, \mathscr{K}^*)$ whose inverse will define D as a principal divisor. □

Corollary 3.4.2

On any scheme X, the map $D \mapsto \mathscr{O}_X(D)$ gives an injective homomorphism of $\mathrm{CaCl}(X)$ to $\mathrm{Pic}(X)$.

Proof By Proposition 3.4.2, we get it immediately. □

Remark The map $\mathrm{CaCl}(X) \to \mathrm{Pic}(X)$ may not be surjective, because there may be invertible sheaves on X which are not isomorphic to any invertible sheaf of $\mathscr{K}$. For an example of Kleiman. On the other hand, this map is an isomorphism in most common situations. Nakai has shown that it is an isomorphism whenever X is projective over a field. We will show now that it is an isomorphism if X is integral.

Proposition 3.4.3

If X is an integral scheme, the homomorphism $\mathrm{CaCl}(X) \to \mathrm{Pic}\, X, [D] \mapsto [\mathscr{O}_X(D)]$ is an isomorphism.

Proof We have only need to show that every invertible sheaf is isomorphic to a subsheaf of $\mathscr{K}$, which in this case is the constant sheaf $K(X)$, where $K(X)$ is the fraction field of X. So let $\mathscr{L}$ be any invertible sheaf, and consider the sheaf $\mathscr{L} \otimes_{\mathscr{O}_X} \mathscr{K}$. On any open set U where $\mathscr{L}|_U \cong \mathscr{O}_U$, we have $\mathscr{L}|_U \otimes_{\mathscr{O}_U} \mathscr{K}|_U \cong \mathscr{K}|_U$, so it is a constant sheaf on U. Now because X is irreducible, it follows that any sheaf whose restriction to each open set of a covering of X is constant, is in fact a constant sheaf. Thus $\mathscr{L} \otimes \mathscr{K}$ is isomorphic to the constant sheaf $\mathscr{K}$, and the natural map $\mathscr{L} \to \mathscr{L} \otimes \mathscr{K} \cong \mathscr{K}$ expresses $\mathscr{L}$ as a subsheaf of $\mathscr{K}$. □

Corollary 3.4.3

If X is a noetherian, integral, separated locally factorial scheme, then there is a natural isomorphism $\mathrm{Cl}(X) \cong \mathrm{Pic}(X)$.

Proof This follows from Proposition 3.4.3 and Proposition 3.4.1. □

Corollary 3.4.4 (Classification of line bundles over $\mathbb{P}_k^n$)

If $X = \mathbb{P}_k^n$ for some field k, then every invertible sheaf on X is isomorphic to $\mathscr{O}_X(l)$ for some $l \in \mathbb{Z}$.

Proof By Theorem 3.2.1, $\mathrm{Cl}(X) \cong \mathbb{Z}$, by Corollary 3.4.3, $\mathrm{Pic}(X) \cong \mathbb{Z}$. Furthermore the generator of $\mathrm{Cl}(X)$ is a hyperplane, which corresponds to the invertible sheaf $\mathscr{O}_X(1)$, by Proposition 1.6.1. Hence $\mathrm{Pic}(X)$ is the free group generated by $\mathscr{O}_X(1)$, and any invertible sheaf $\mathscr{L}$ is isomorphic to $\mathscr{O}_X(l)$ for some $l \in \mathbb{Z}$. □

3.4.4 Associated closed subscheme of codimension 1 of effective Cartier divisor

We conclude this section with some remarks about closed subschemes of codimension one of a scheme X.

Definition 3.4.4 (Associated closed subscheme of codimension 1 of effective Cartier divisor)

A Cartier divisor on a scheme X is effective if can be represented by $\{(U_i, f_i)\}$ where all the $f_i \in \Gamma(U_i, \mathscr{O}_{U_i})$ (see Definition 3.4.2). In that case we define the associated subscheme of codimension 1, Y, to be the closed subscheme defined by the sheaf of ideals $\mathscr{I}$ which is locally generated by f_i.

Remark Clearly this gives a $1-1$ correspondence between effective Cartier divisors on X and **locally principal closed subschemes** Y, i.e., subschemes whose sheaf of ideals is locally generated by a single element. Note also that if X is an integral separated noetherian locally factorial scheme, so that the Cartier divisors correspond to Weil divisors by Proposition 3.4.1, then the effective Cartier divisors correspond exactly to the effective Weil divisors.

Proposition 3.4.4

Let D be an effective Cartier divisor on a scheme X, and let Y be the associated locally principal closed subscheme. Then $\mathscr{I}_Y \cong \mathscr{O}_X(-D)$.

Proof Let $D = \{(U_i, f_i)\}$, then $\mathscr{O}_X(-D)$ is the subsheaf of $\mathscr{K}$ generated locally by f_i. Since D is effect, then $f_i \in \mathscr{O}_X(U_i)$, and therefore $\mathscr{O}_X(-D)|_{U_i} = f_i\mathscr{O}_{U_i} \subseteq \mathscr{O}_{U_i}$, so $\mathscr{I}_Y|_{U_i} = \mathscr{O}_X(-D)|_{U_i}$. Hence $\mathscr{I}_Y \cong \mathscr{O}_X(-D)$. $\square$

Part II

Ample Line Bundles and Linear Series

Chapter 4 Ample and Nef Line Bundles

Chapter 5 Linear Series

Chapter 6 geometric Consequence of Positivity

Chapter 7   Local Positivity

Part III

Positivity for Vector Bundles

Part IV

Multiplier Ideals and Their Applications

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